

BFS Traversal

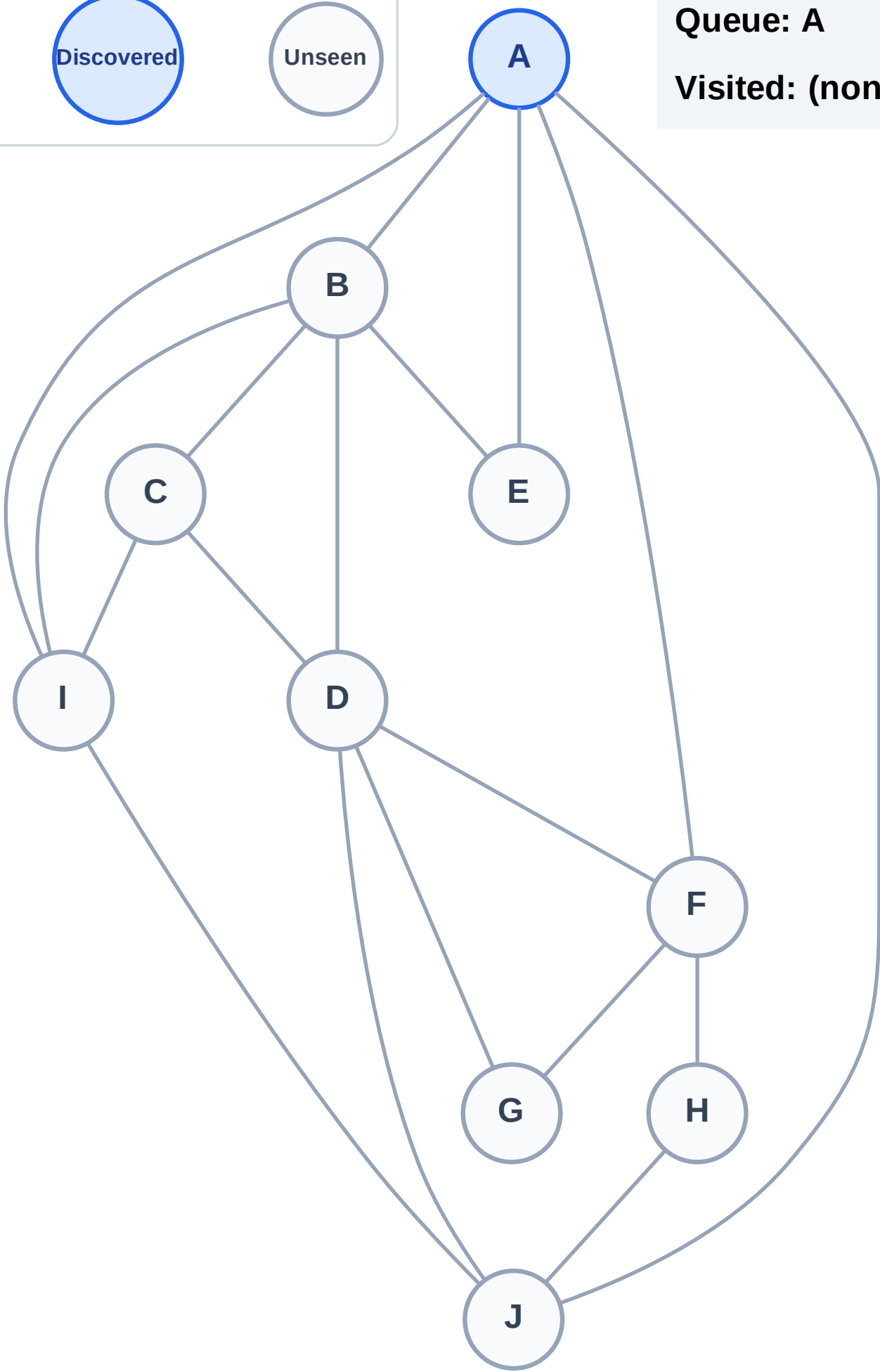
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

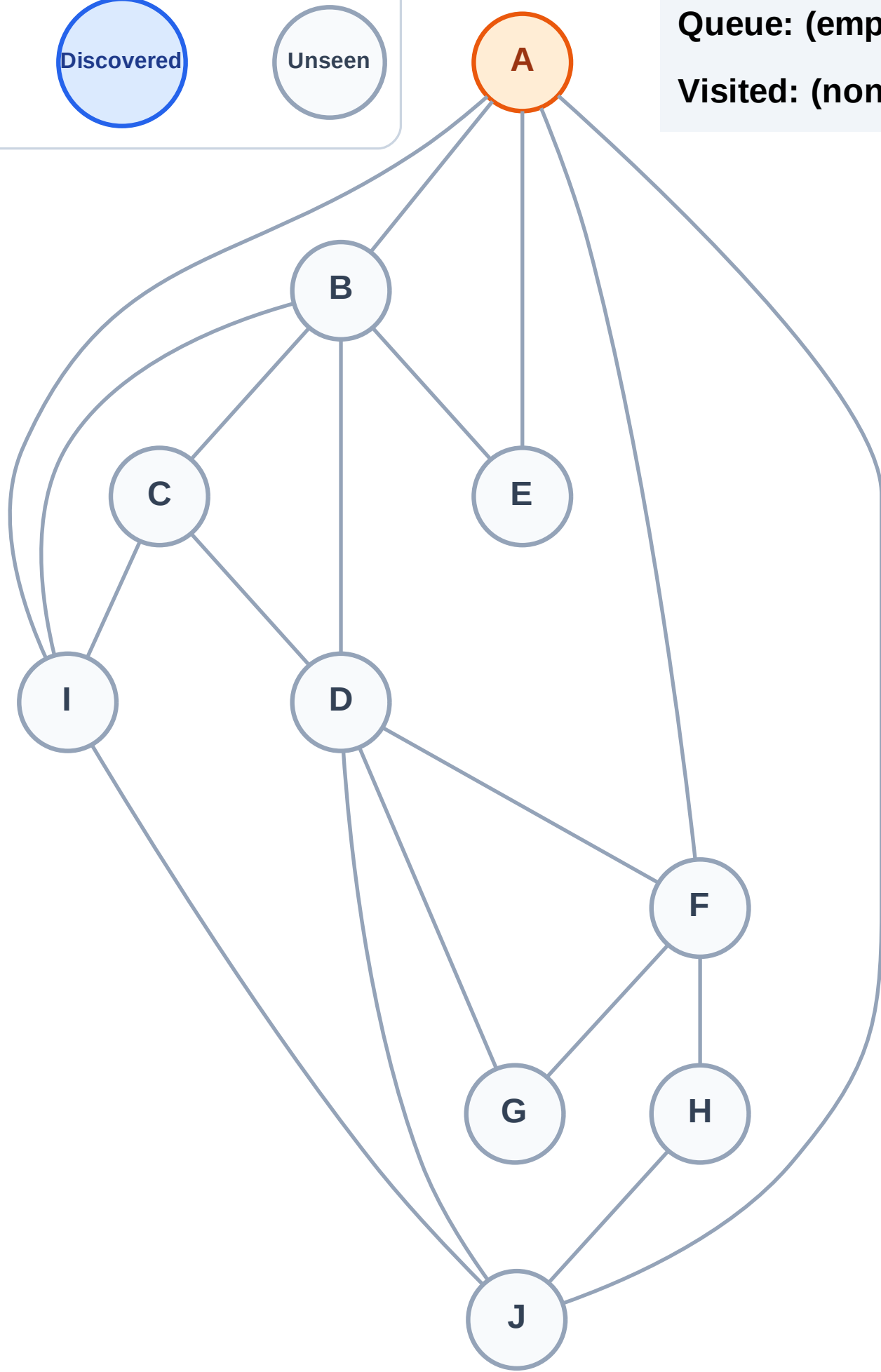
Step 2: Process node A

Legend



Queue: (empty)

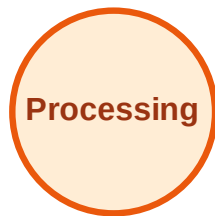
Visited: (none)



BFS Traversal

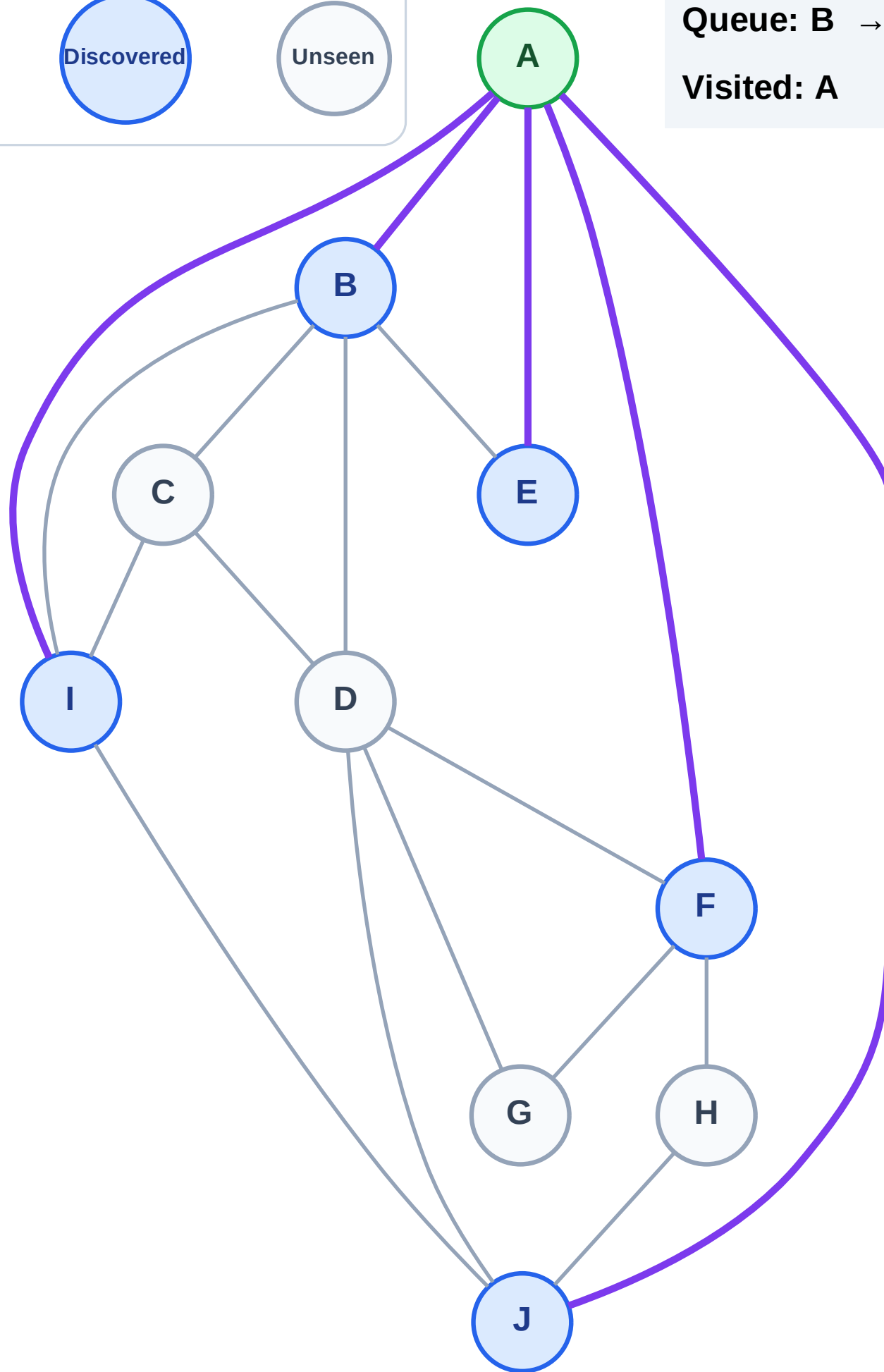
Step 3: Discover B, E, F, I, J from A

Legend



Queue: B → E → F → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

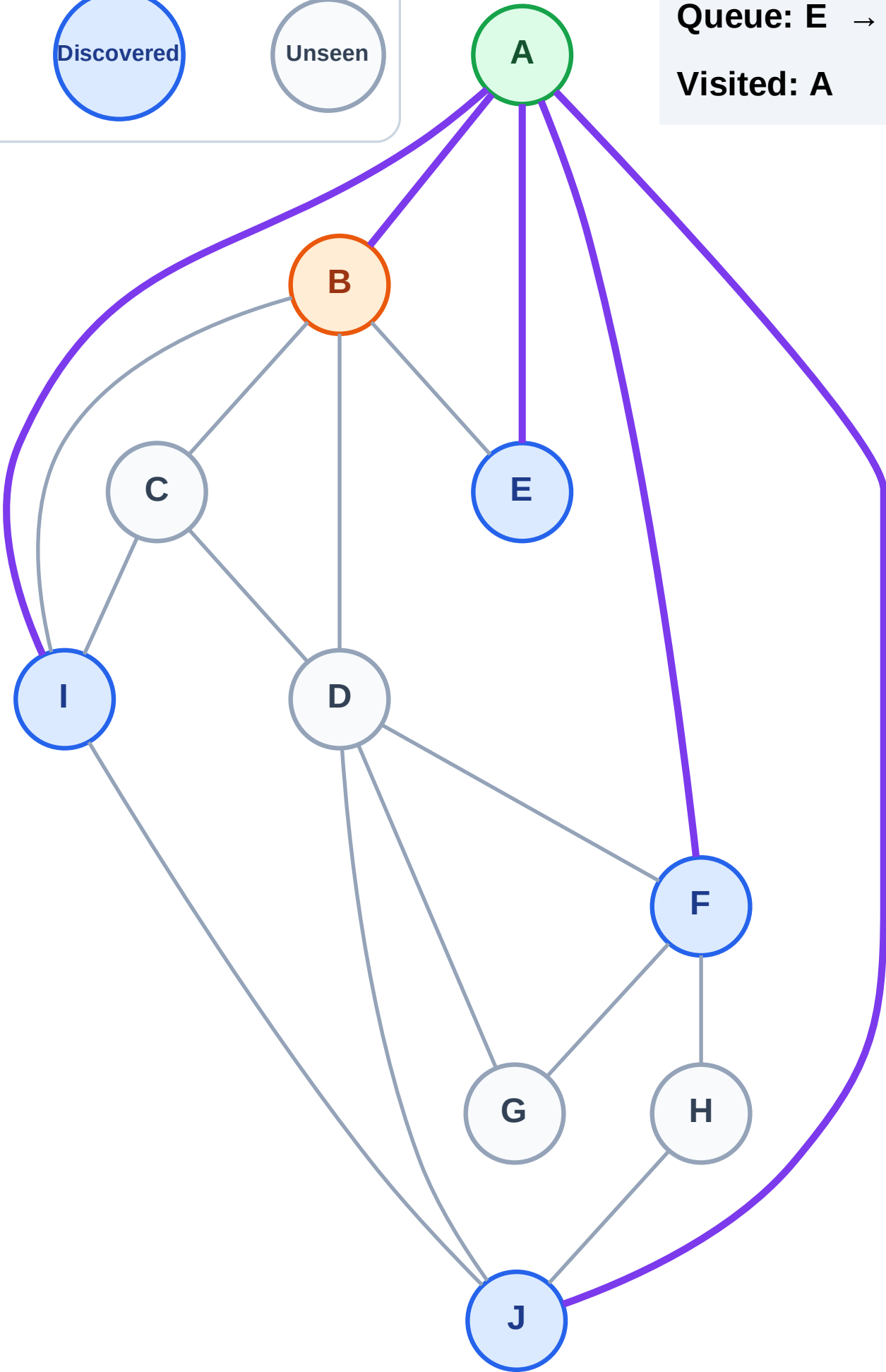
Processing

Discovered

Unseen

Queue: E → F → I → J

Visited: A



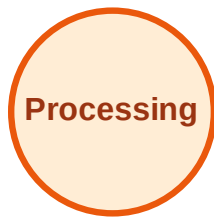
BFS Traversal

Step 5: Discover C, D from B

Legend



Complete



Processing



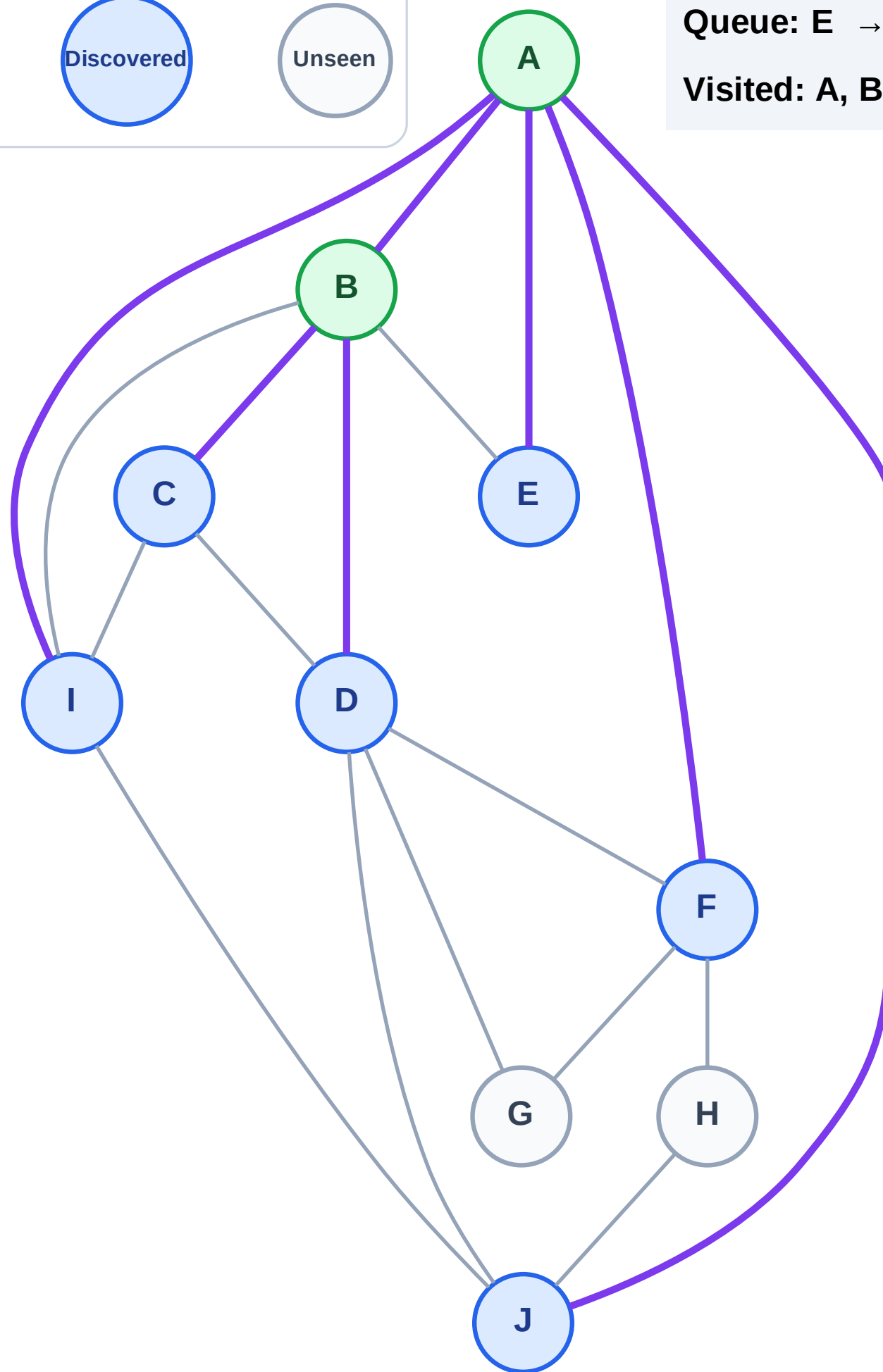
Discovered



Unseen

Queue: E → F → I → J → C → D

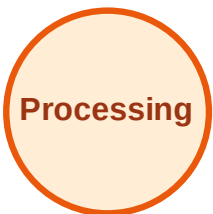
Visited: A, B



BFS Traversal

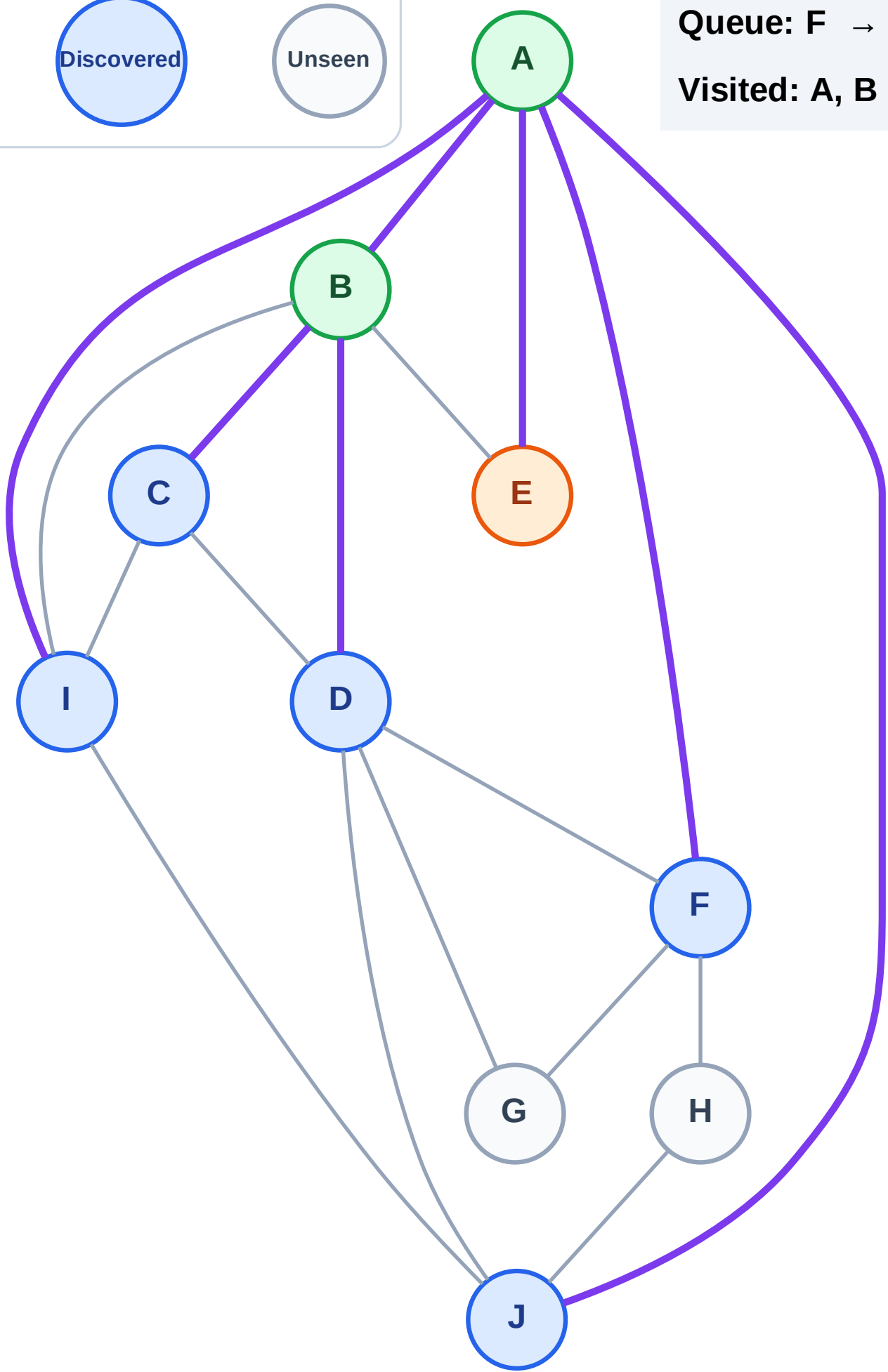
Step 6: Process node E

Legend



Queue: F → I → J → C → D

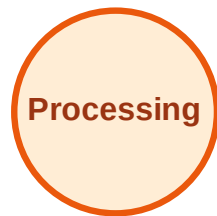
Visited: A, B



BFS Traversal

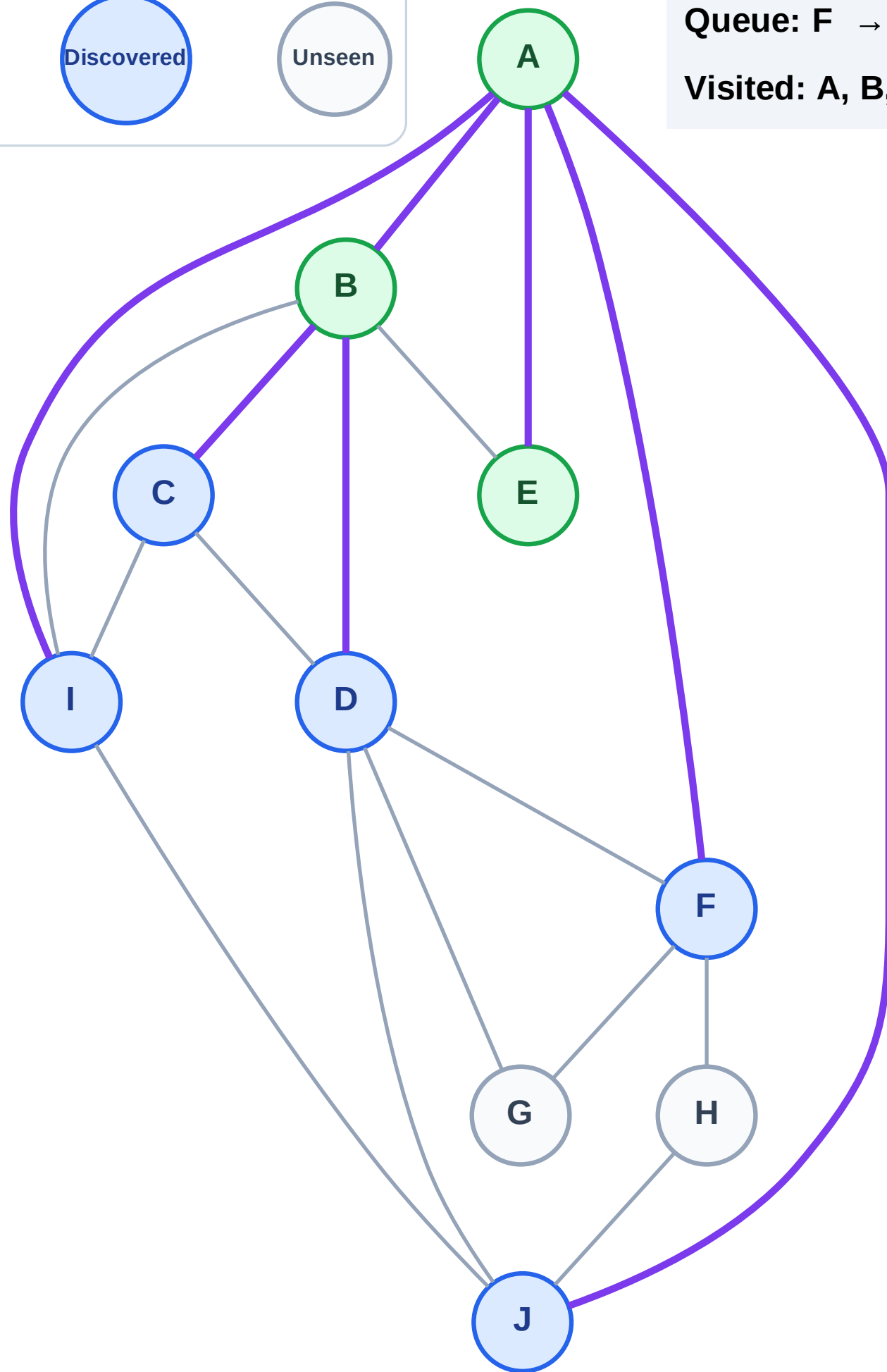
Step 7: No new neighbors from E

Legend



Queue: F → I → J → C → D

Visited: A, B, E



BFS Traversal

Step 8: Process node F

Legend

Complete

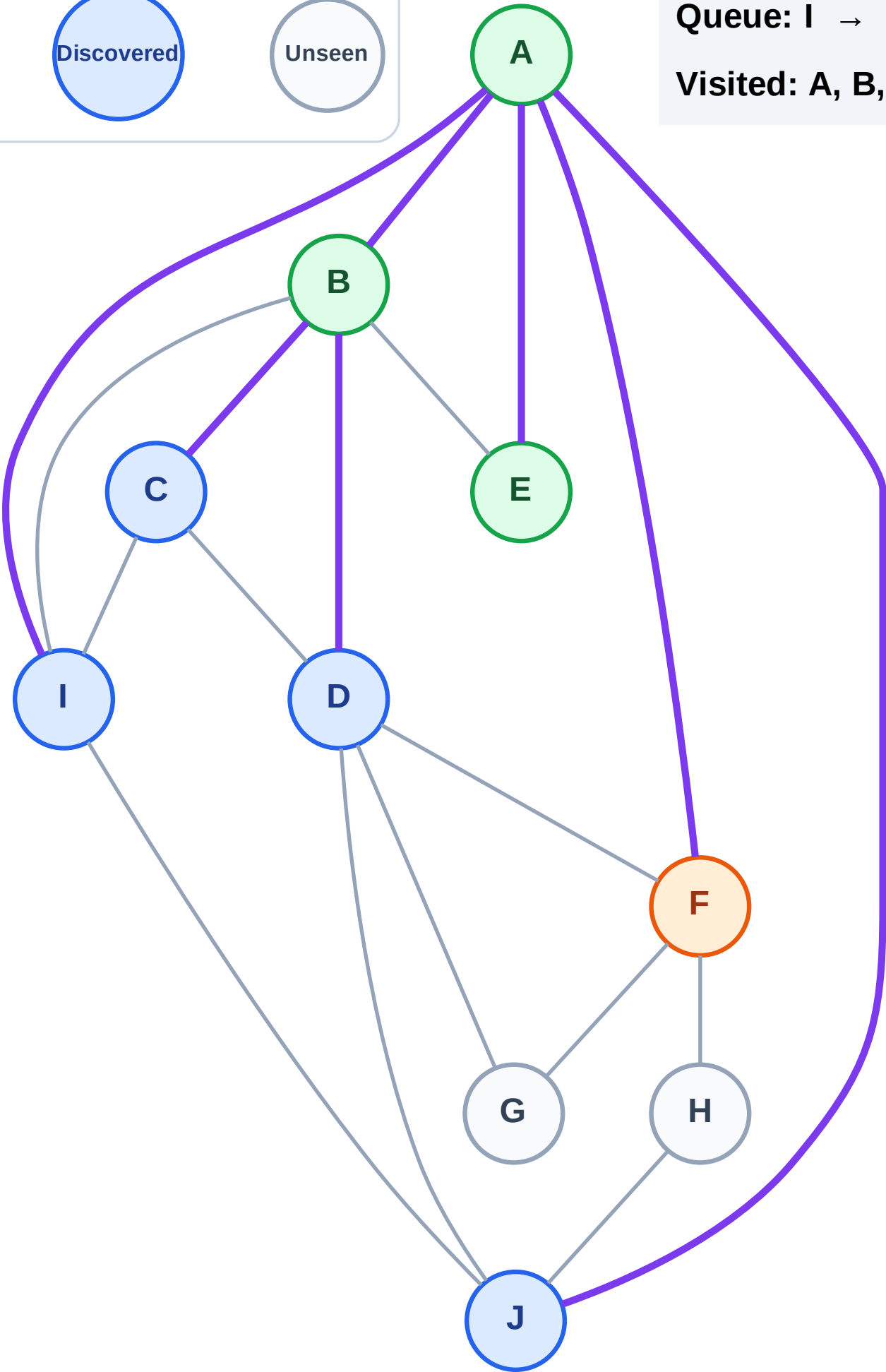
Processing

Discovered

Unseen

Queue: I → J → C → D

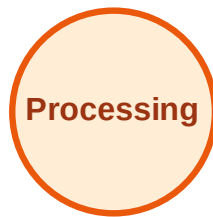
Visited: A, B, E



BFS Traversal

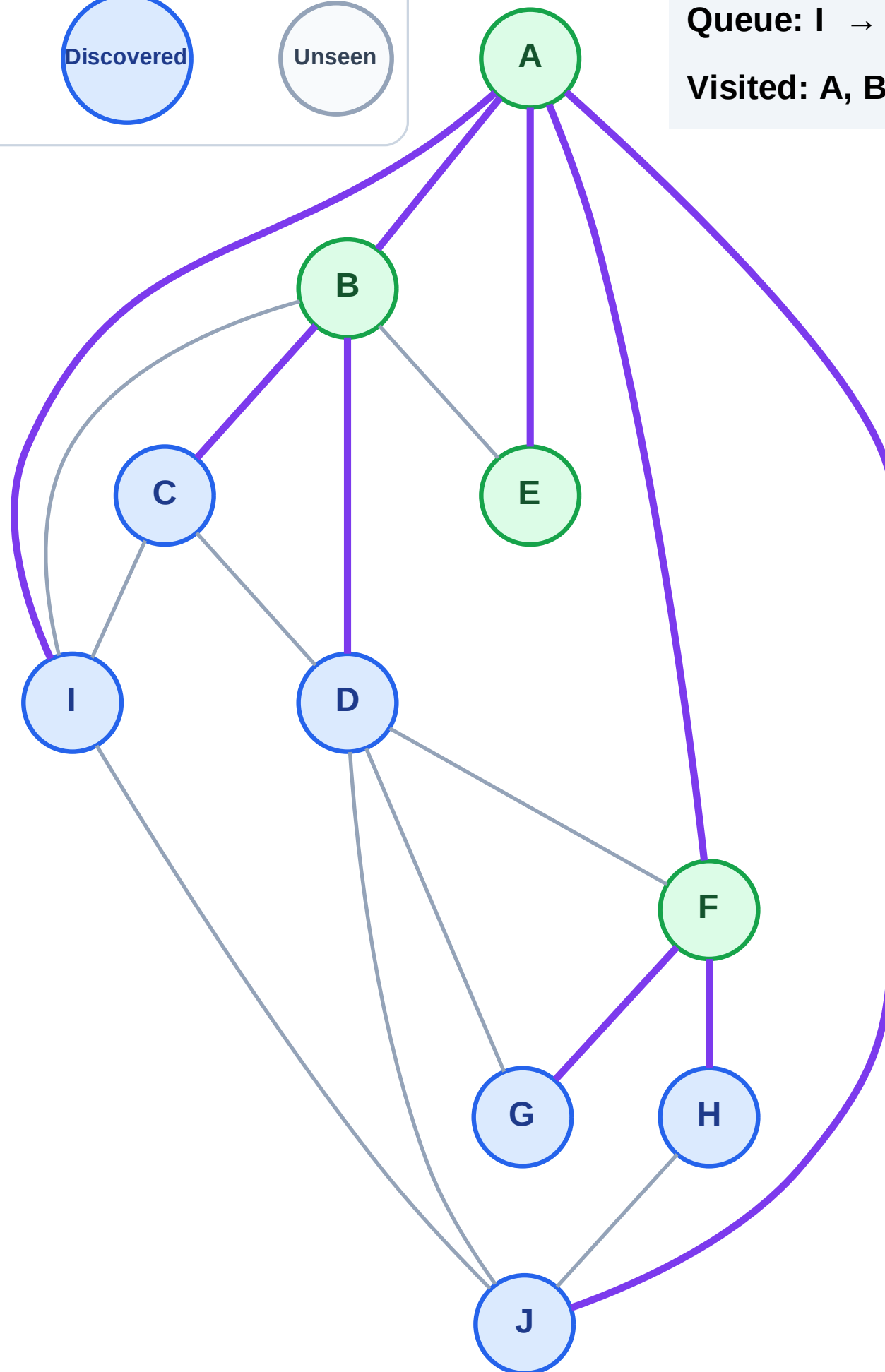
Step 9: Discover G, H from F

Legend



Queue: I → J → C → D → G → H

Visited: A, B, E, F



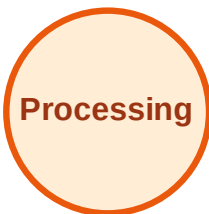
BFS Traversal

Step 10: Process node I

Legend



Complete



Processing



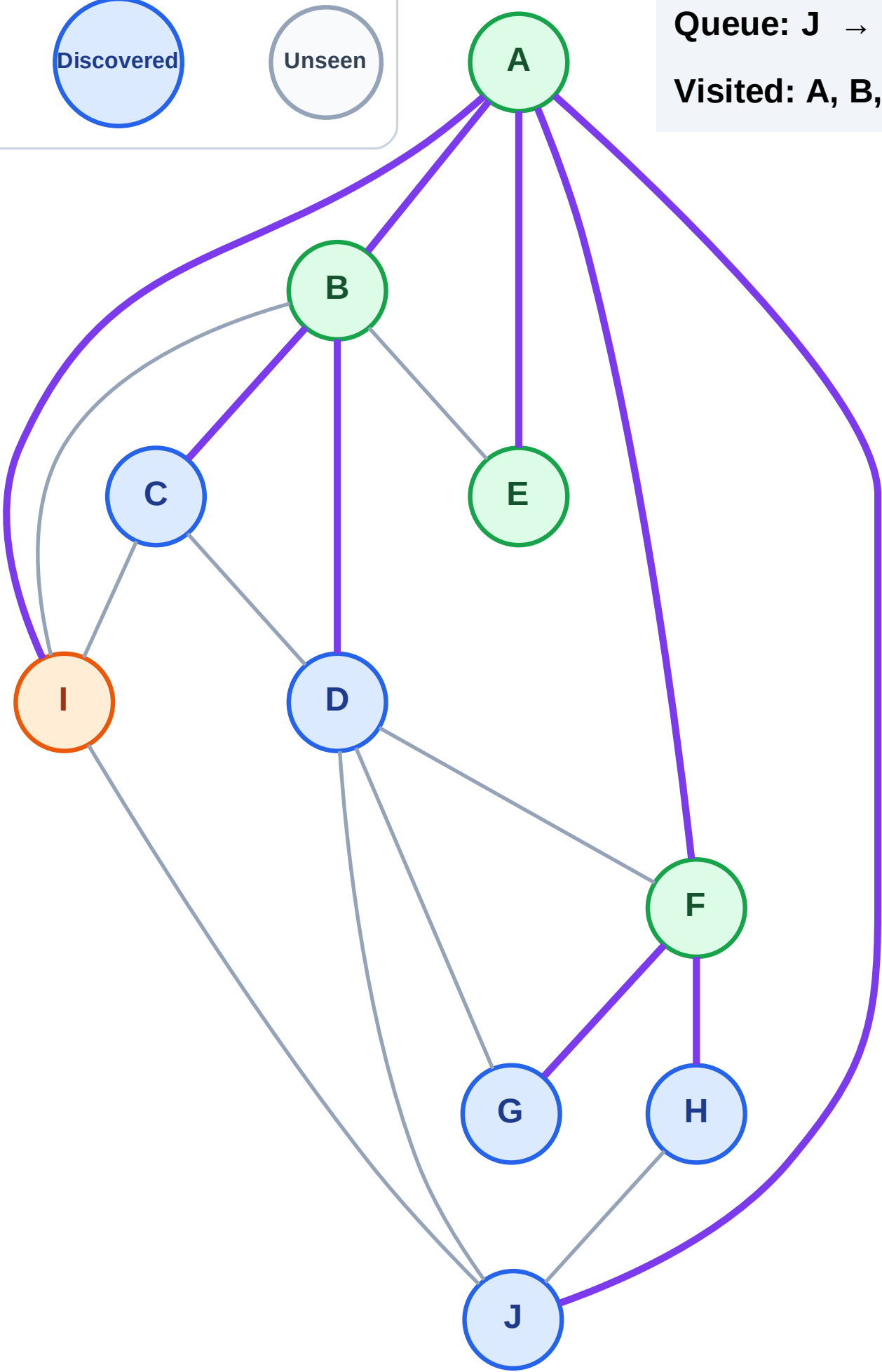
Discovered



Unseen

Queue: J → C → D → G → H

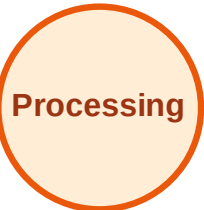
Visited: A, B, E, F



BFS Traversal

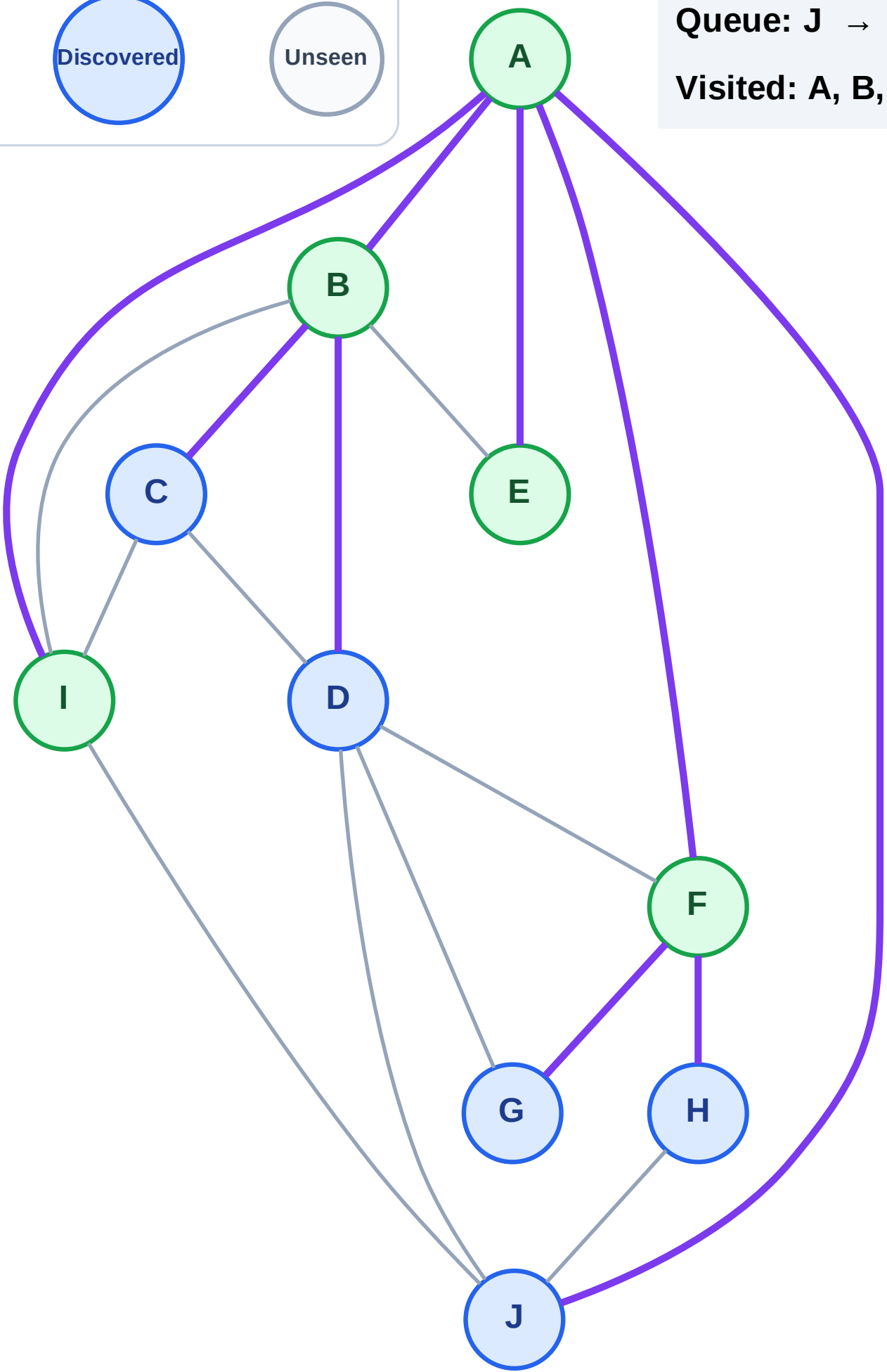
Step 11: No new neighbors from I

Legend



Queue: J → C → D → G → H

Visited: A, B, E, F, I



BFS Traversal

Step 12: Process node J

Legend

Complete

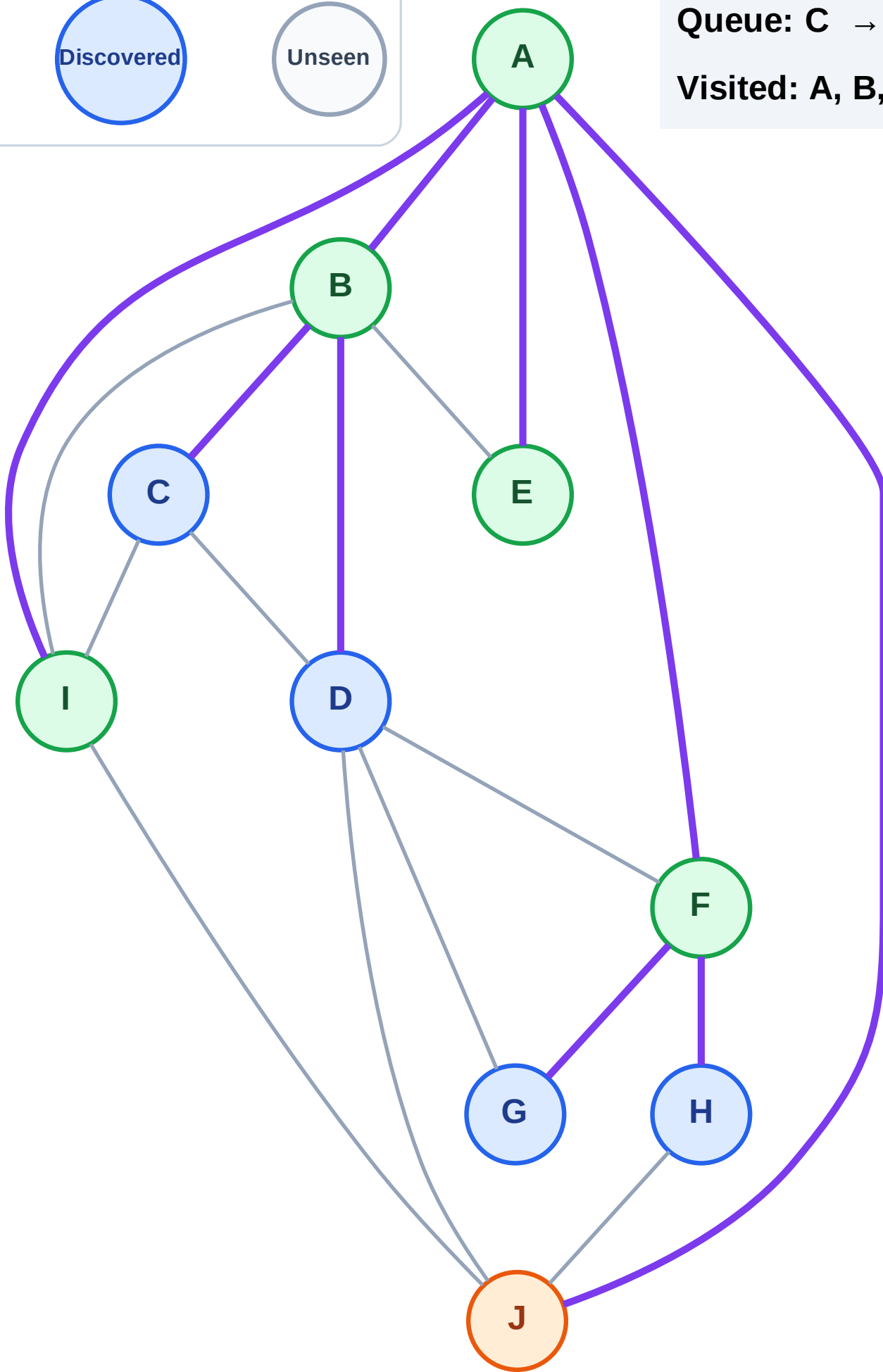
Processing

Discovered

Unseen

Queue: C → D → G → H

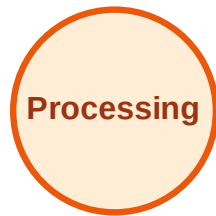
Visited: A, B, E, F, I



BFS Traversal

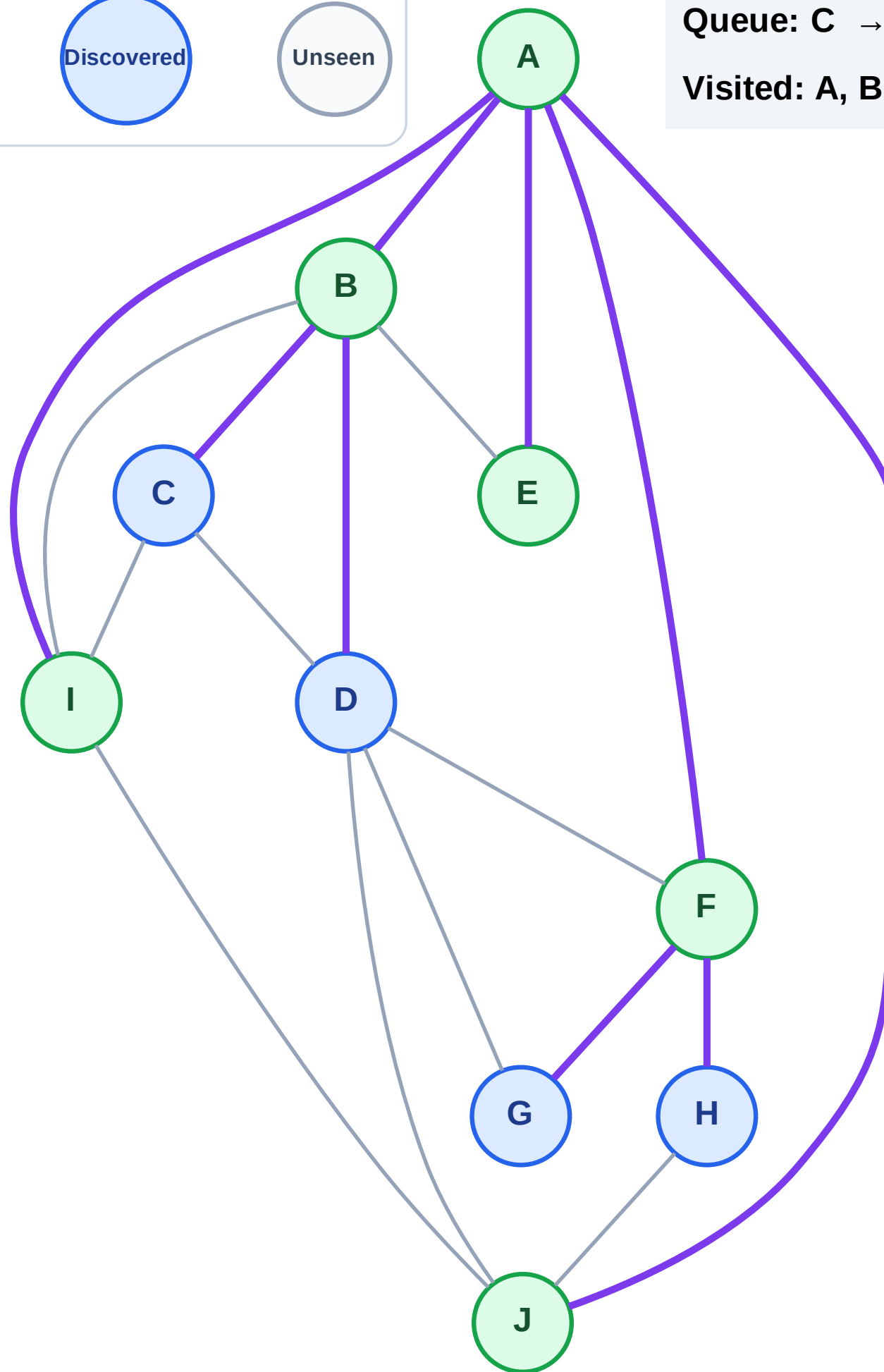
Step 13: No new neighbors from J

Legend



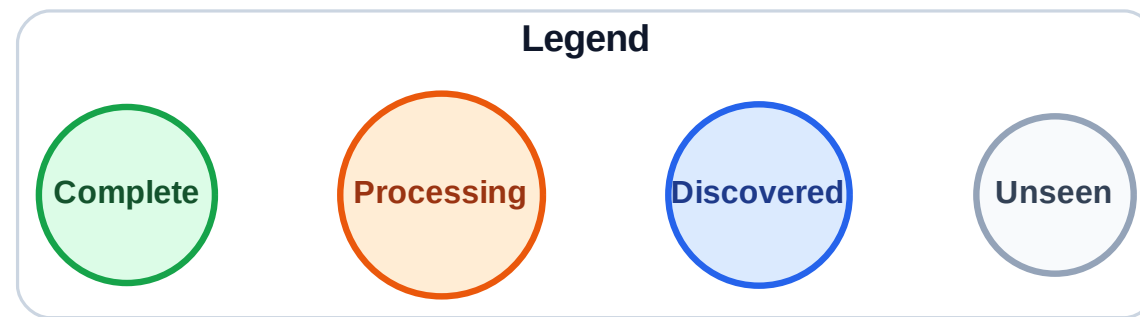
Queue: C → D → G → H

Visited: A, B, E, F, I, J

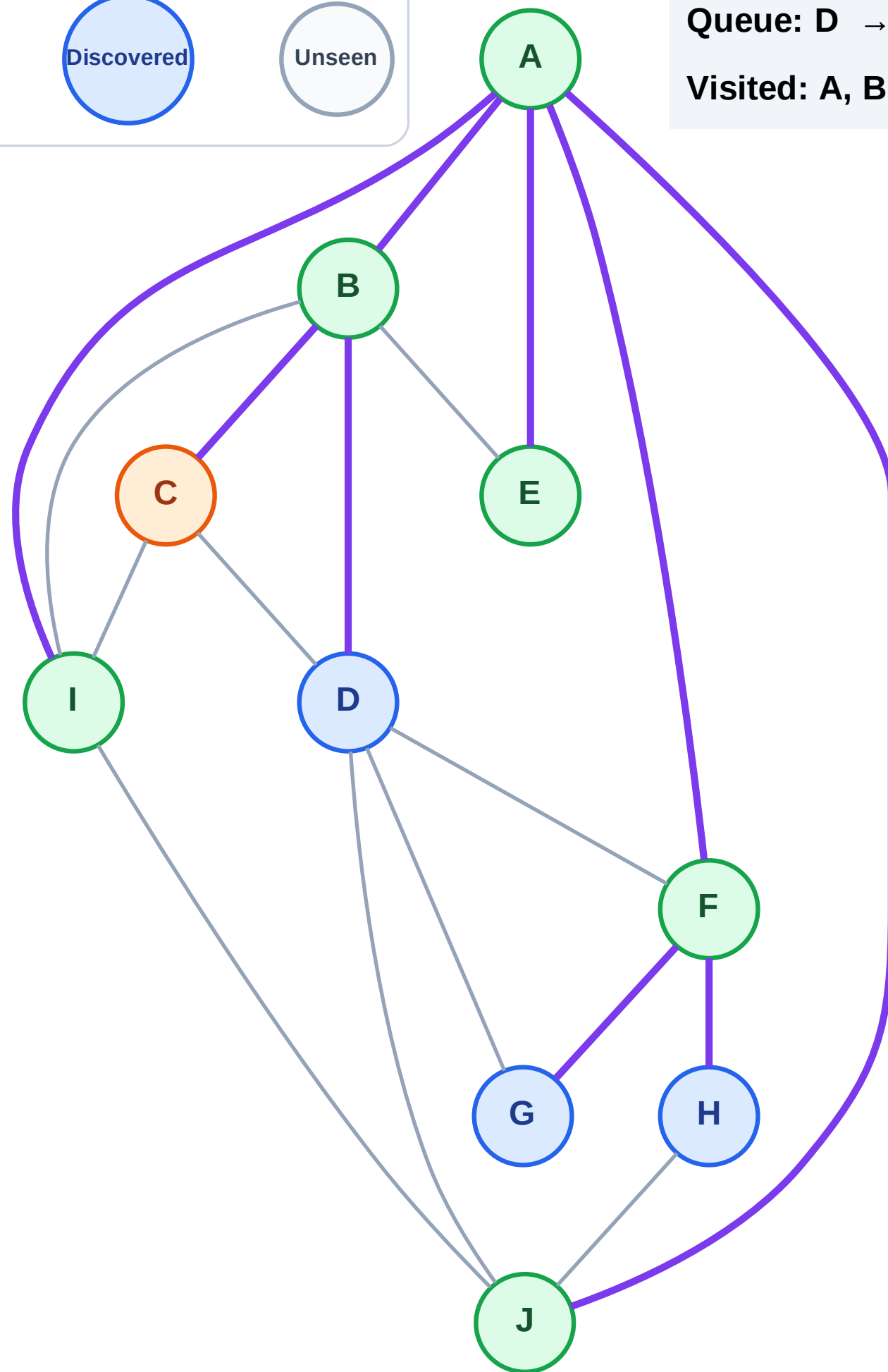


BFS Traversal

Step 14: Process node C



Queue: D → G → H
Visited: A, B, E, F, I, J



BFS Traversal

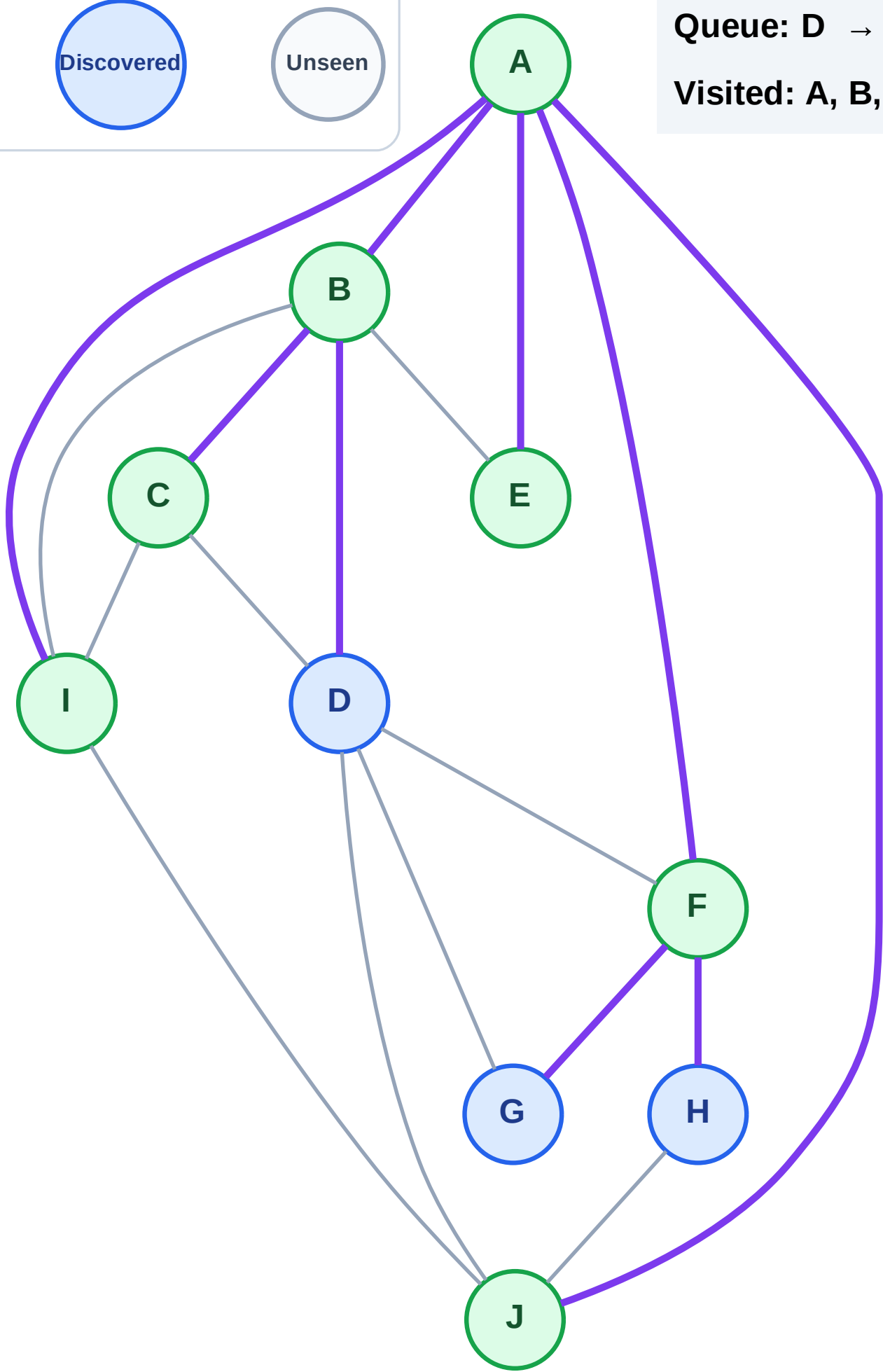
Step 15: No new neighbors from C

Legend



Queue: D → G → H

Visited: A, B, C, E, F, I, J



BFS Traversal

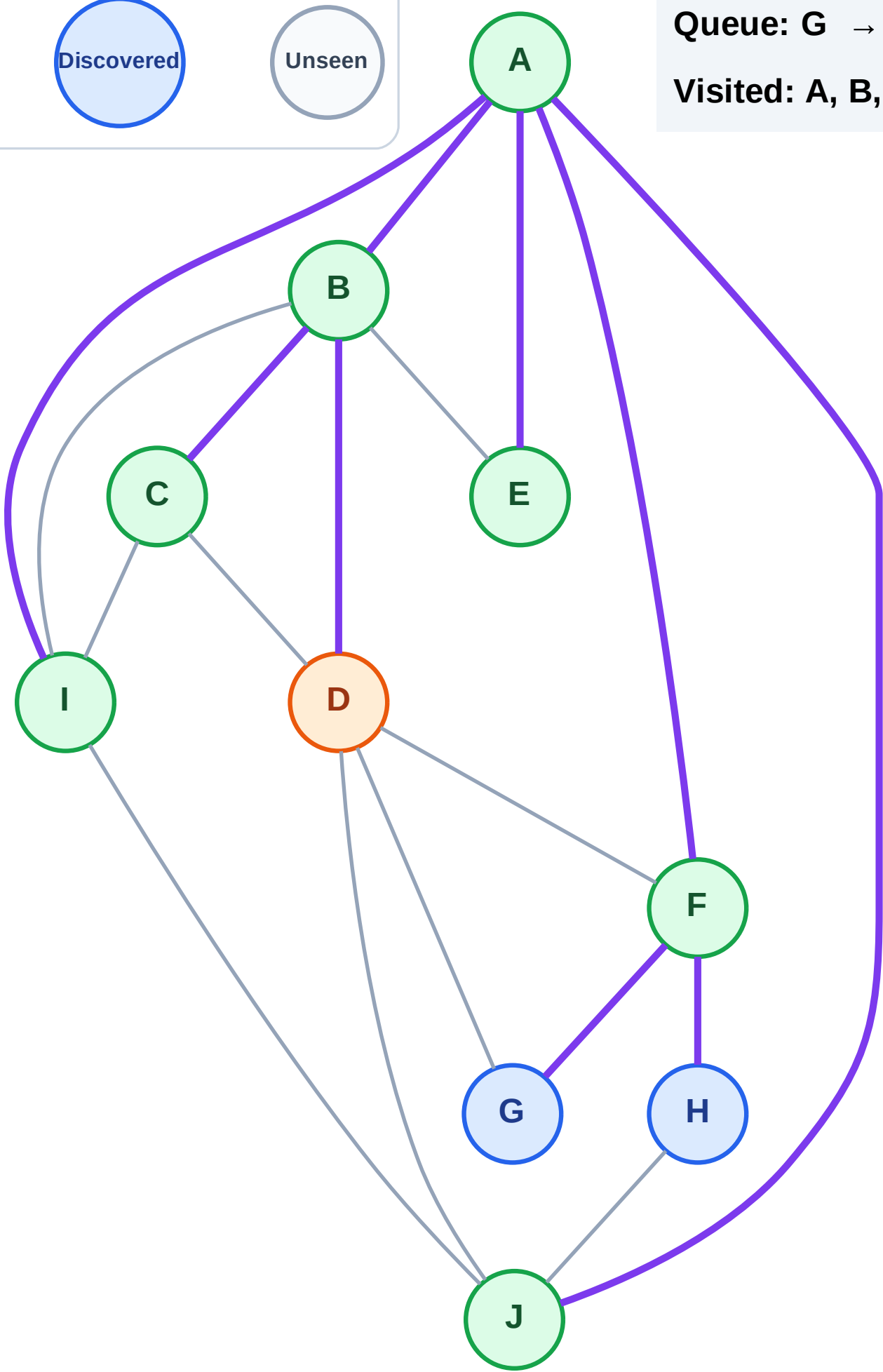
Step 16: Process node D

Legend



Queue: G → H

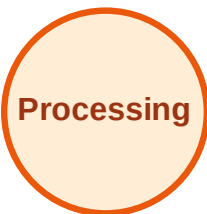
Visited: A, B, C, E, F, I, J



BFS Traversal

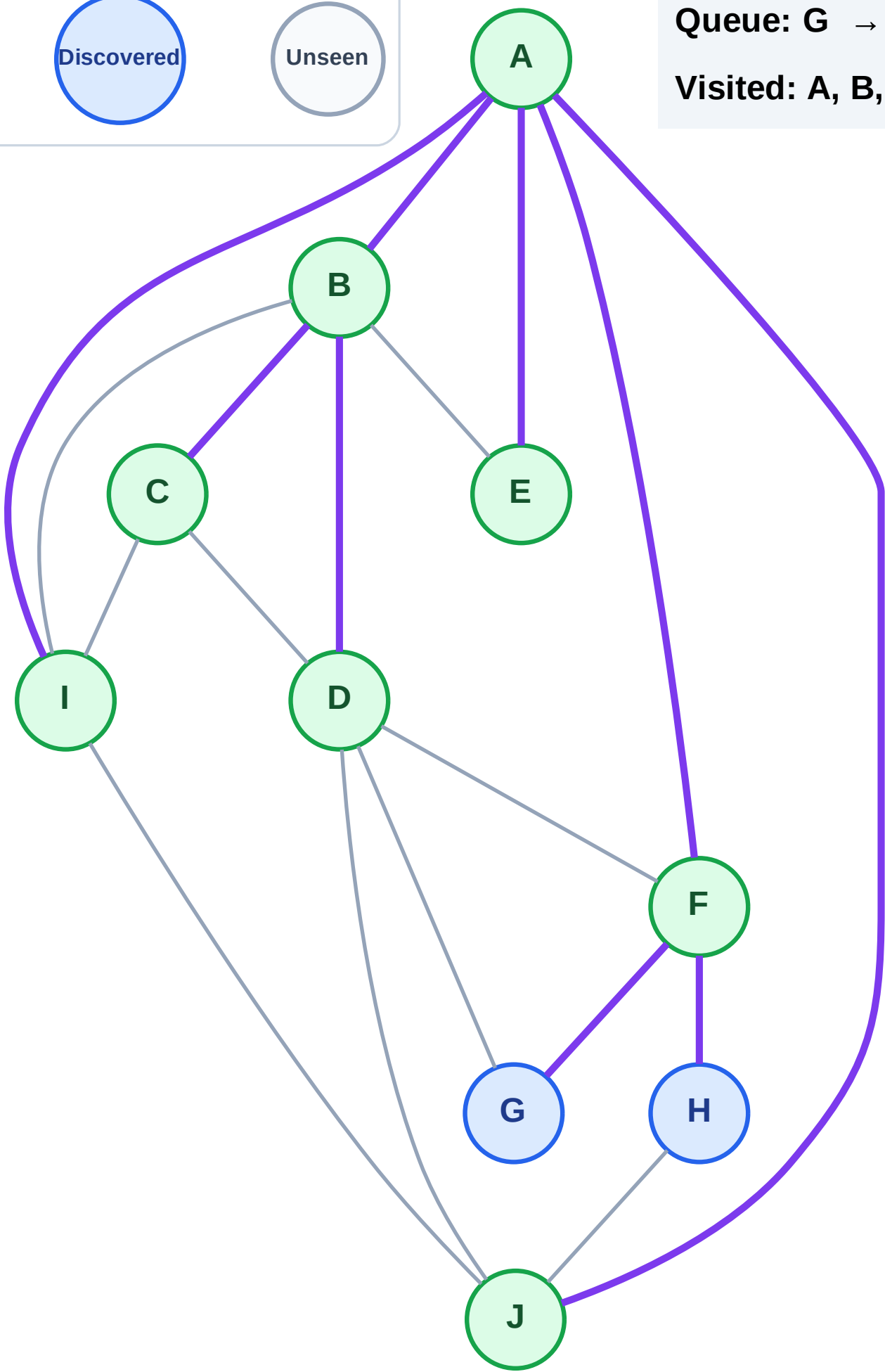
Step 17: No new neighbors from D

Legend



Queue: G → H

Visited: A, B, C, D, E, F, I, J



BFS Traversal

Step 18: Process node G

Legend

Complete

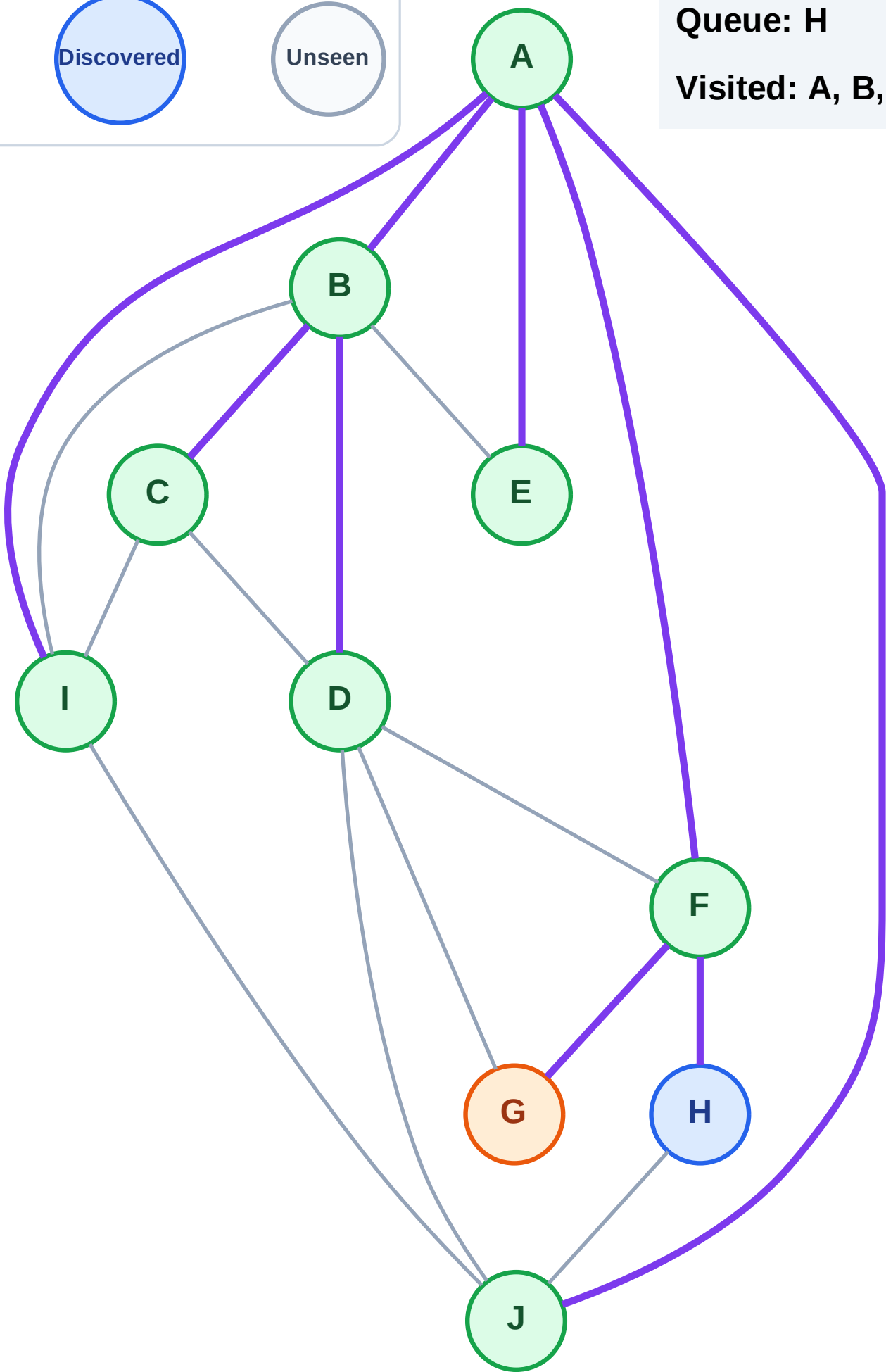
Processing

Discovered

Unseen

Queue: H

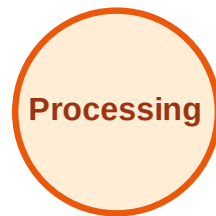
Visited: A, B, C, D, E, F, I, J



BFS Traversal

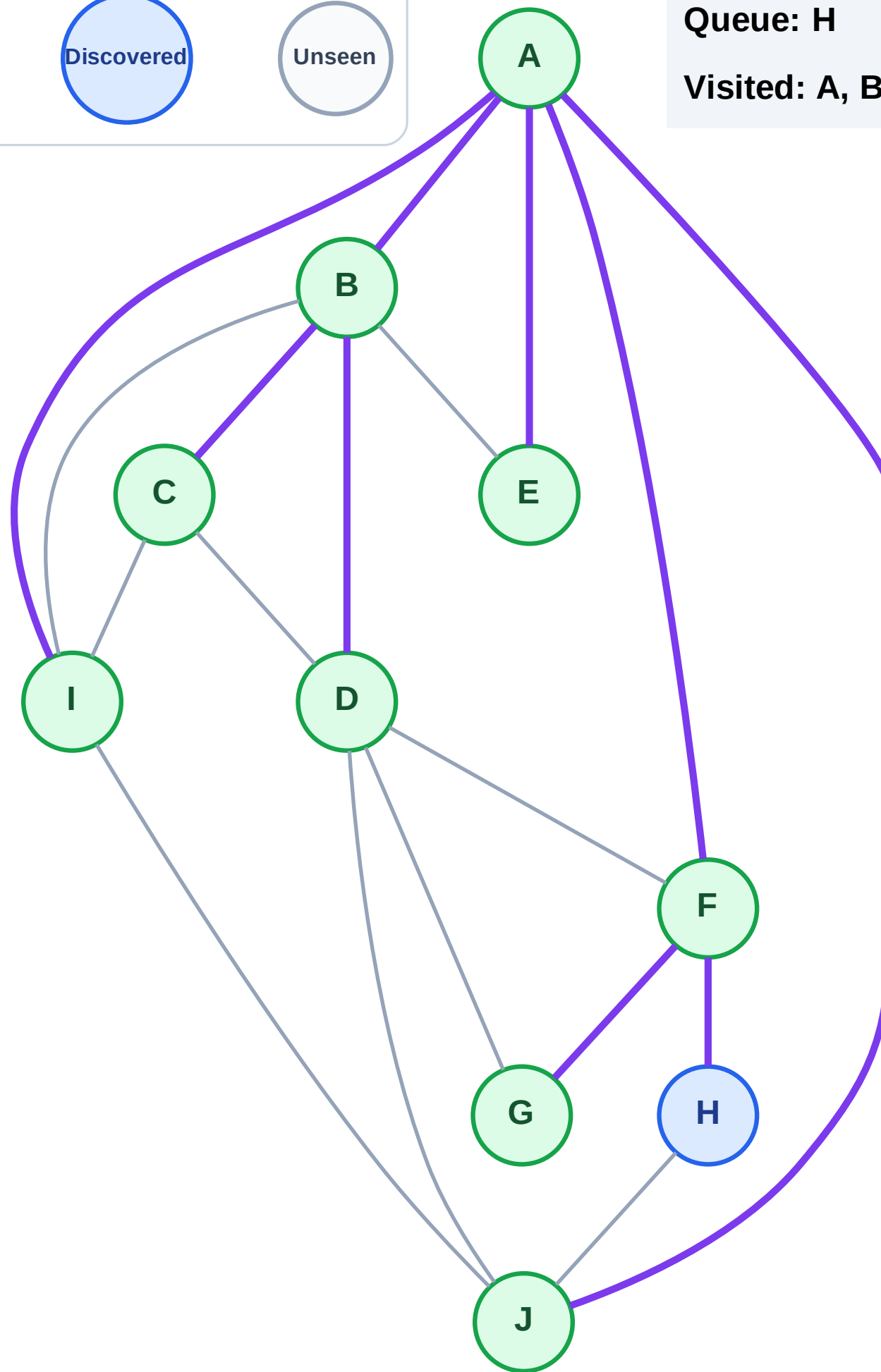
Step 19: No new neighbors from G

Legend



Queue: H

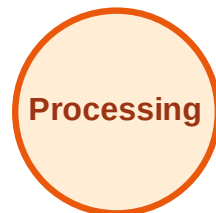
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

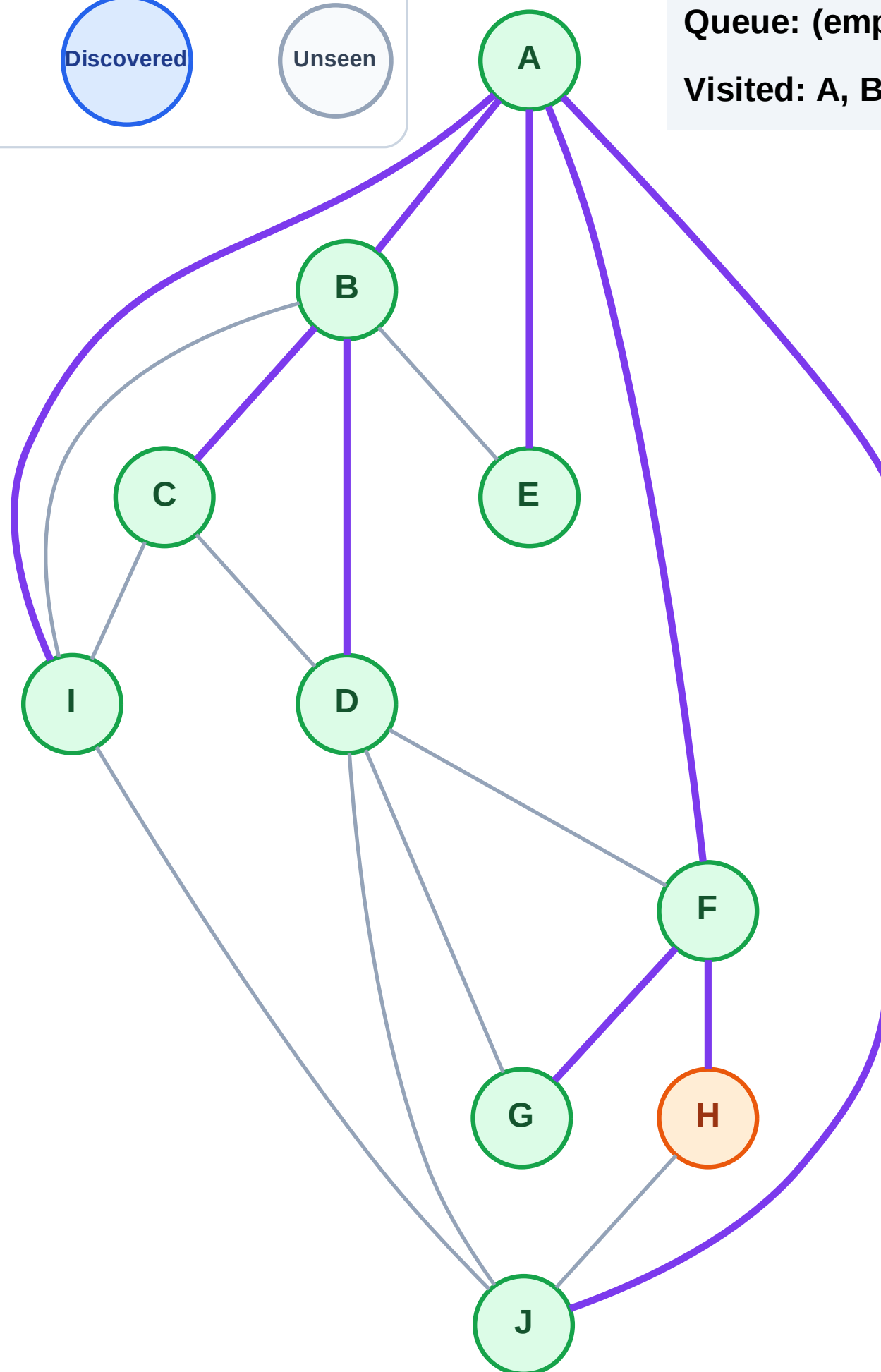
Step 20: Process node H

Legend



Queue: (empty)

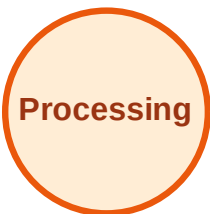
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

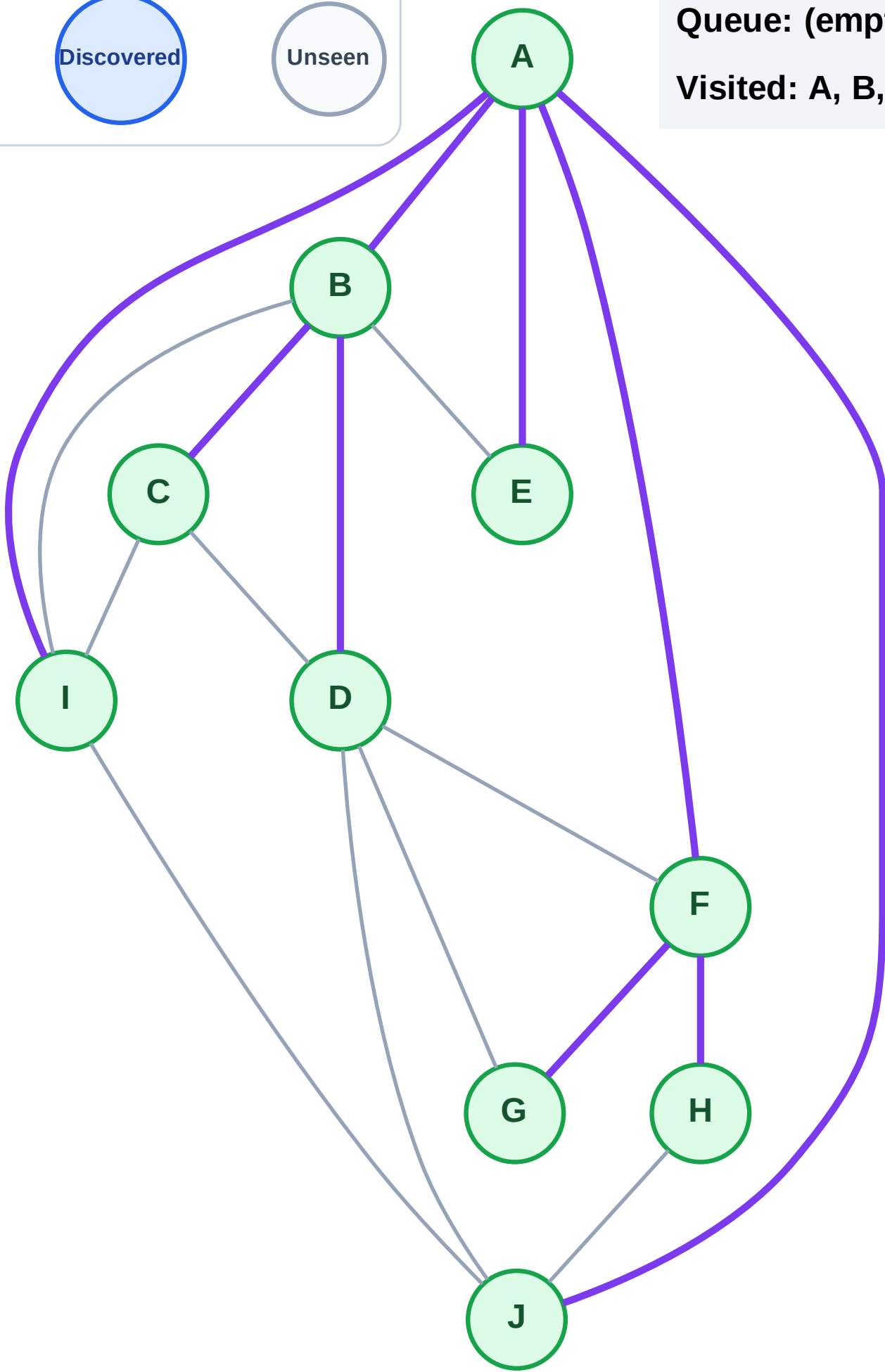
Step 21: No new neighbors from H

Legend



Queue: (empty)

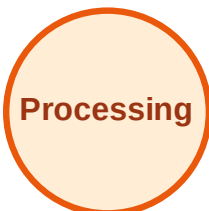
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

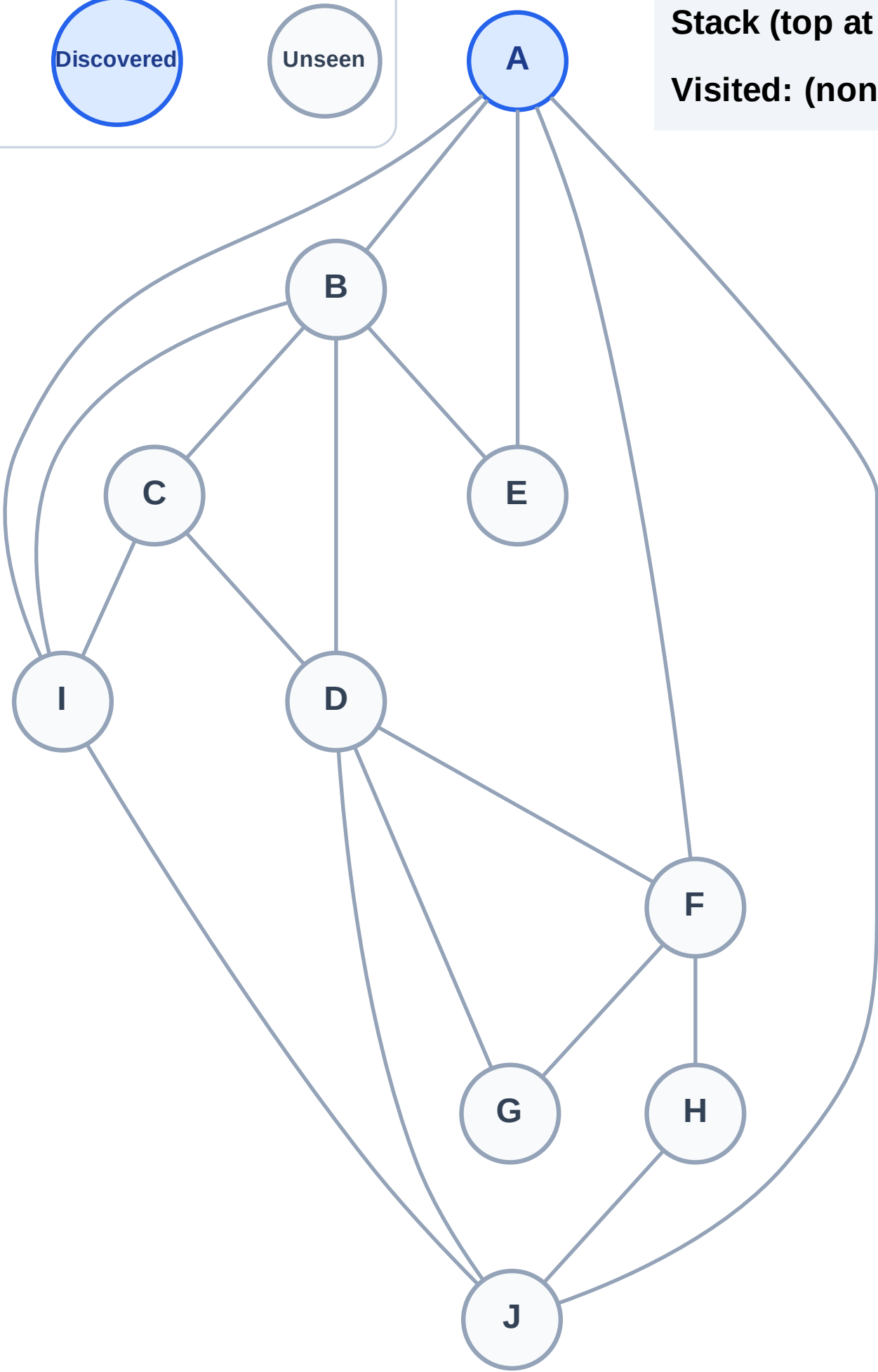
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

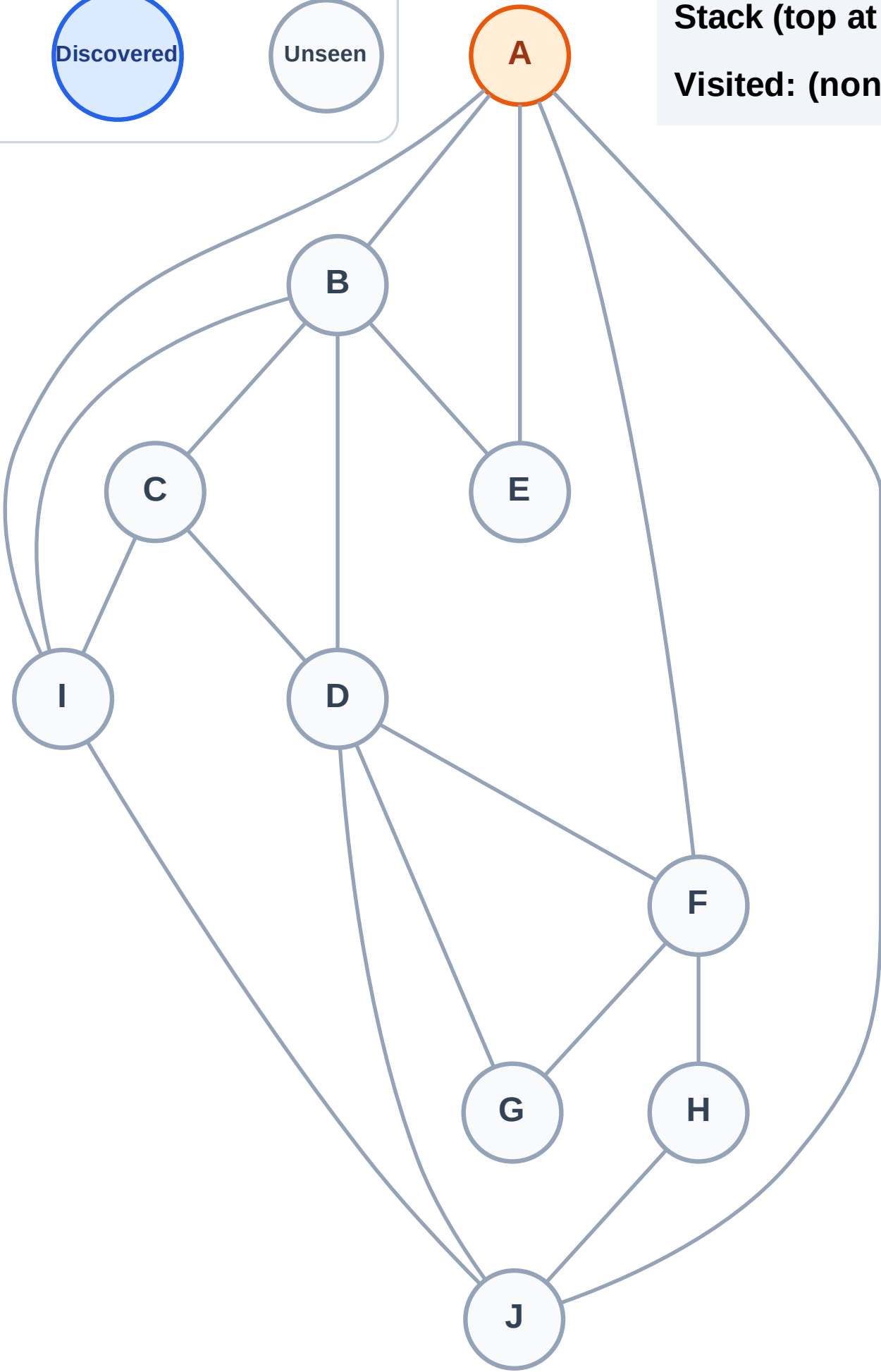
Complete

Processing

Discovered

Unseen

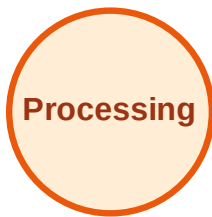
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

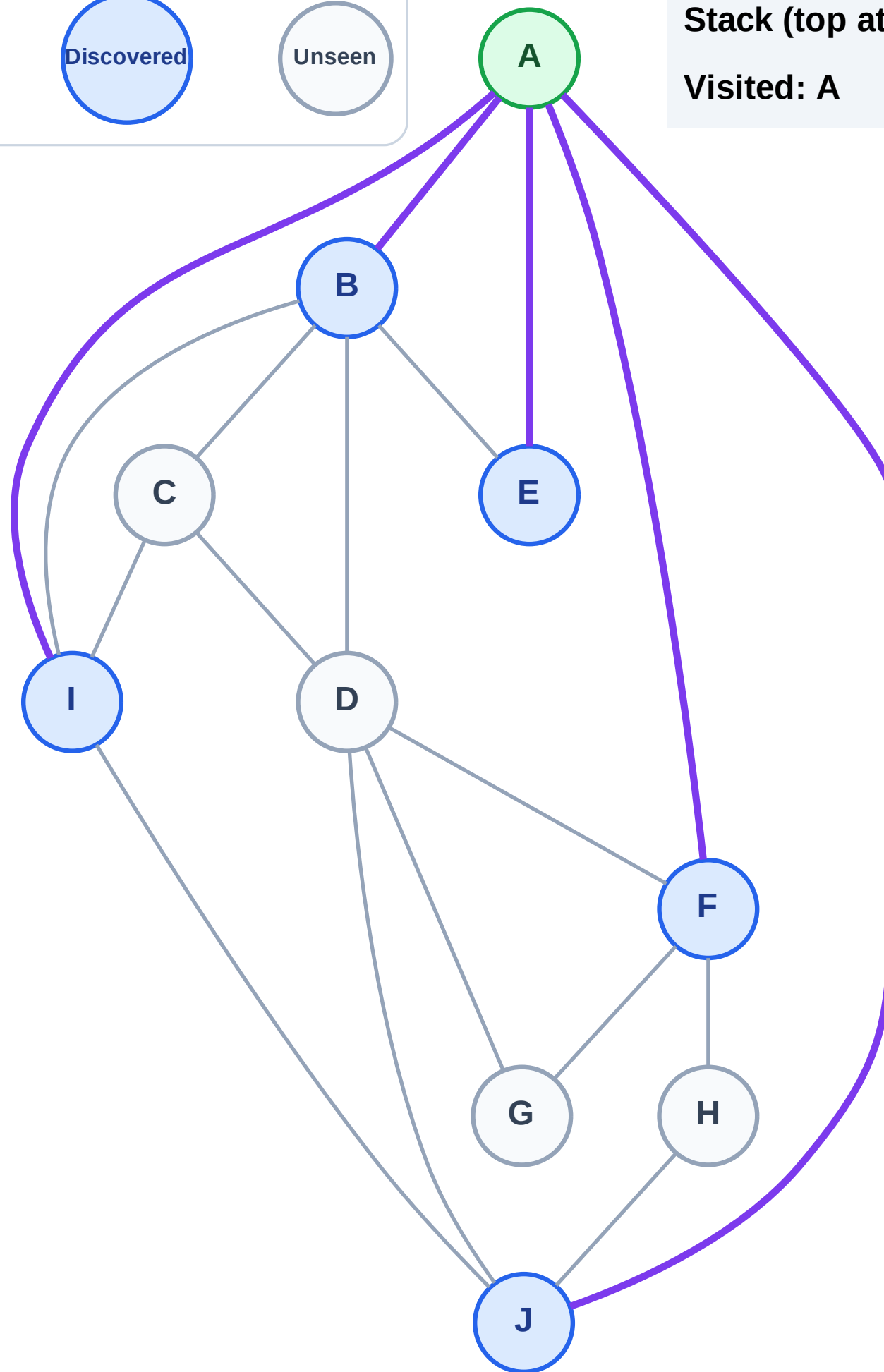
Step 3: Discover J, I, F, E, B from A

Legend



Stack (top at right): J | I | F | E | B

Visited: A



DFS Traversal

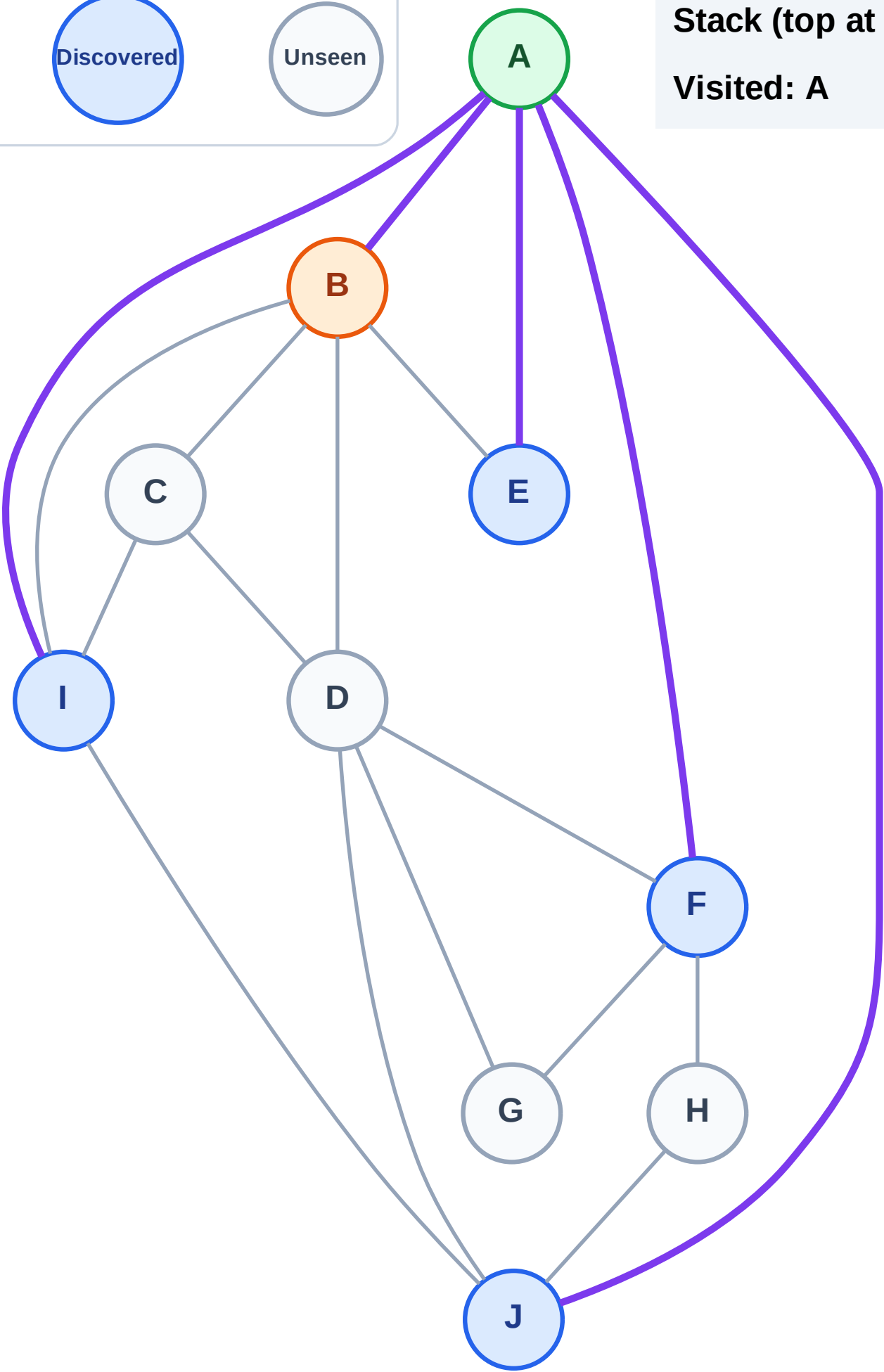
Step 4: Process node B

Legend



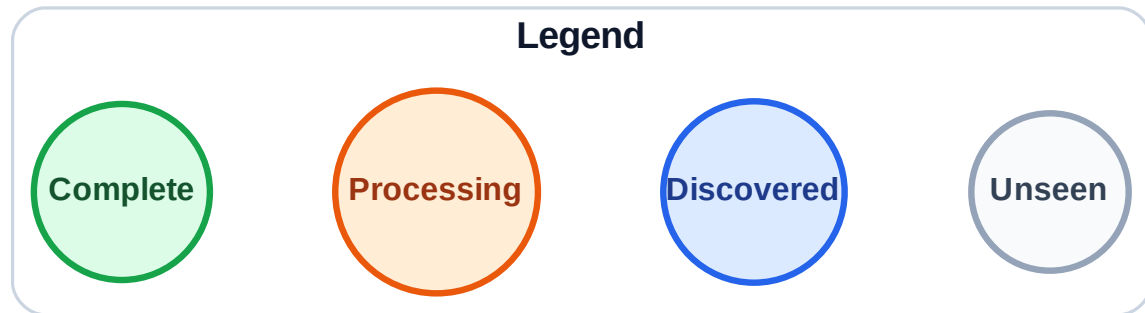
Stack (top at right): J | I | F | E

Visited: A



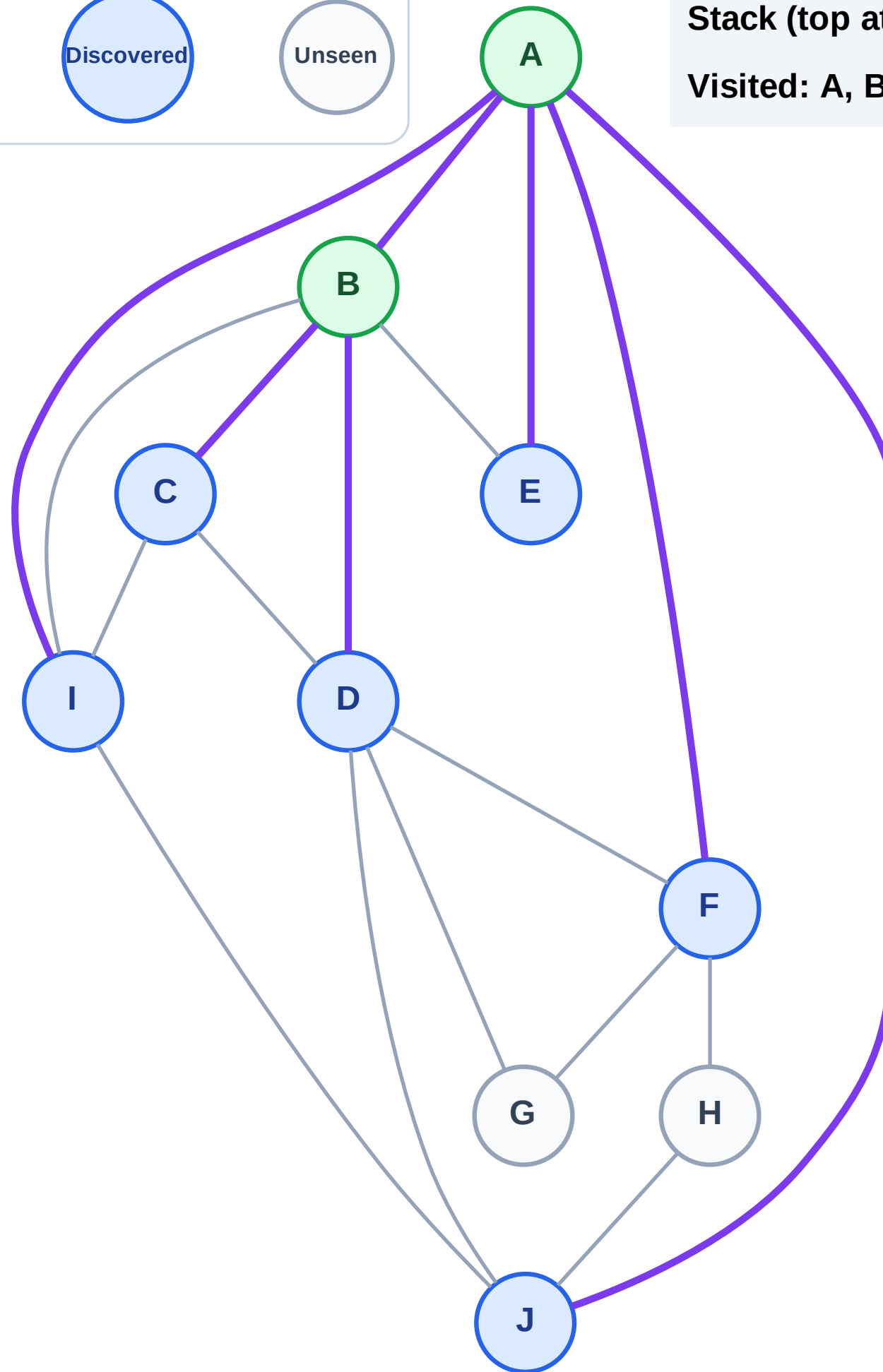
Step 5: Discover D, C from B

Step 5: Discover D, C from B



Stack (top at right): J | I | F | E | D | C

Visited: A, B



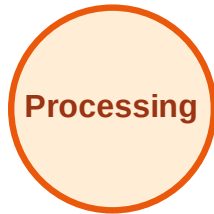
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



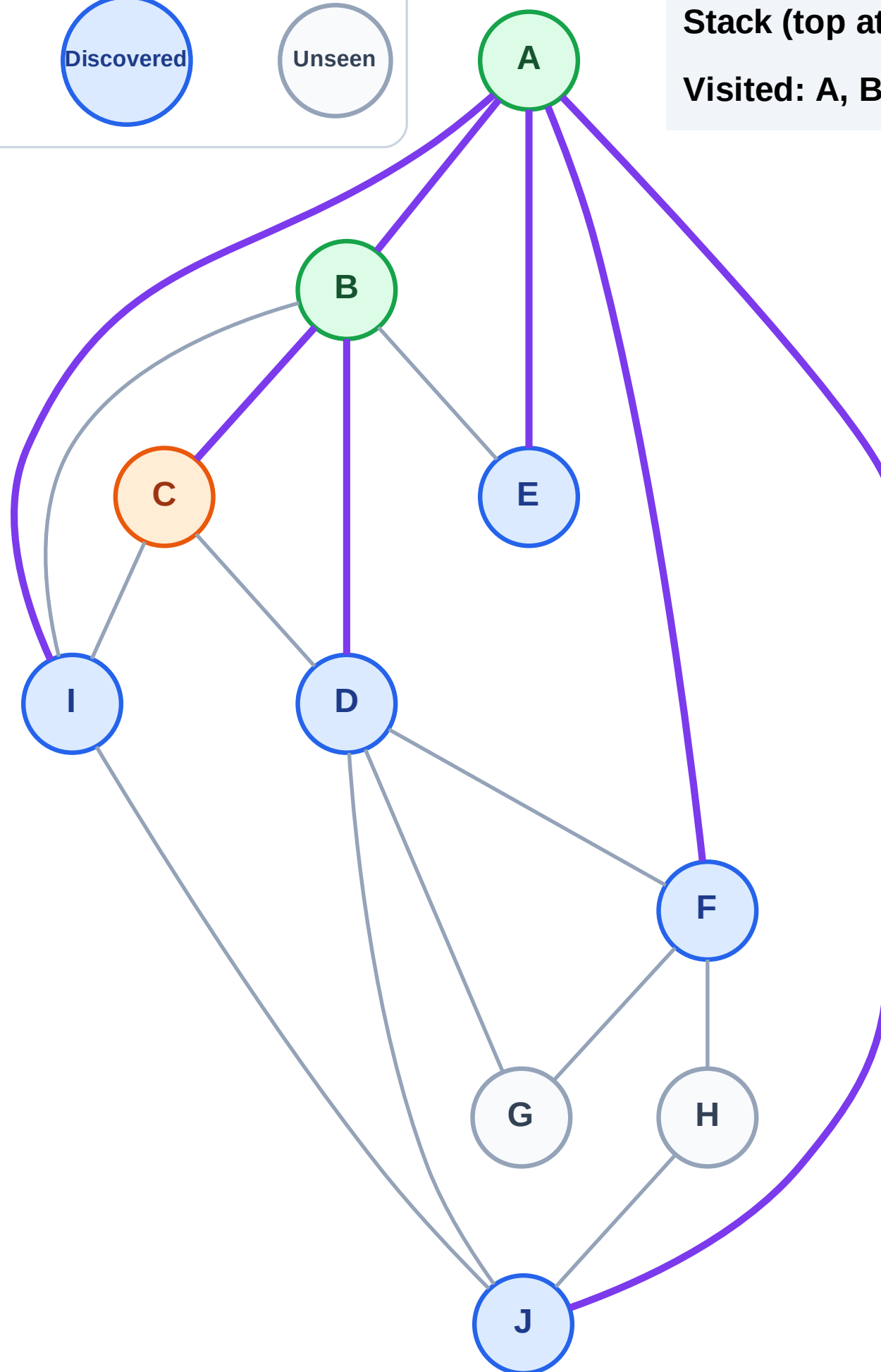
Discovered



Unseen

Stack (top at right): J | I | F | E | D

Visited: A, B



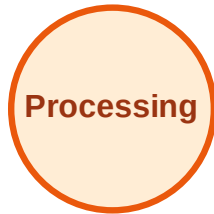
DFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



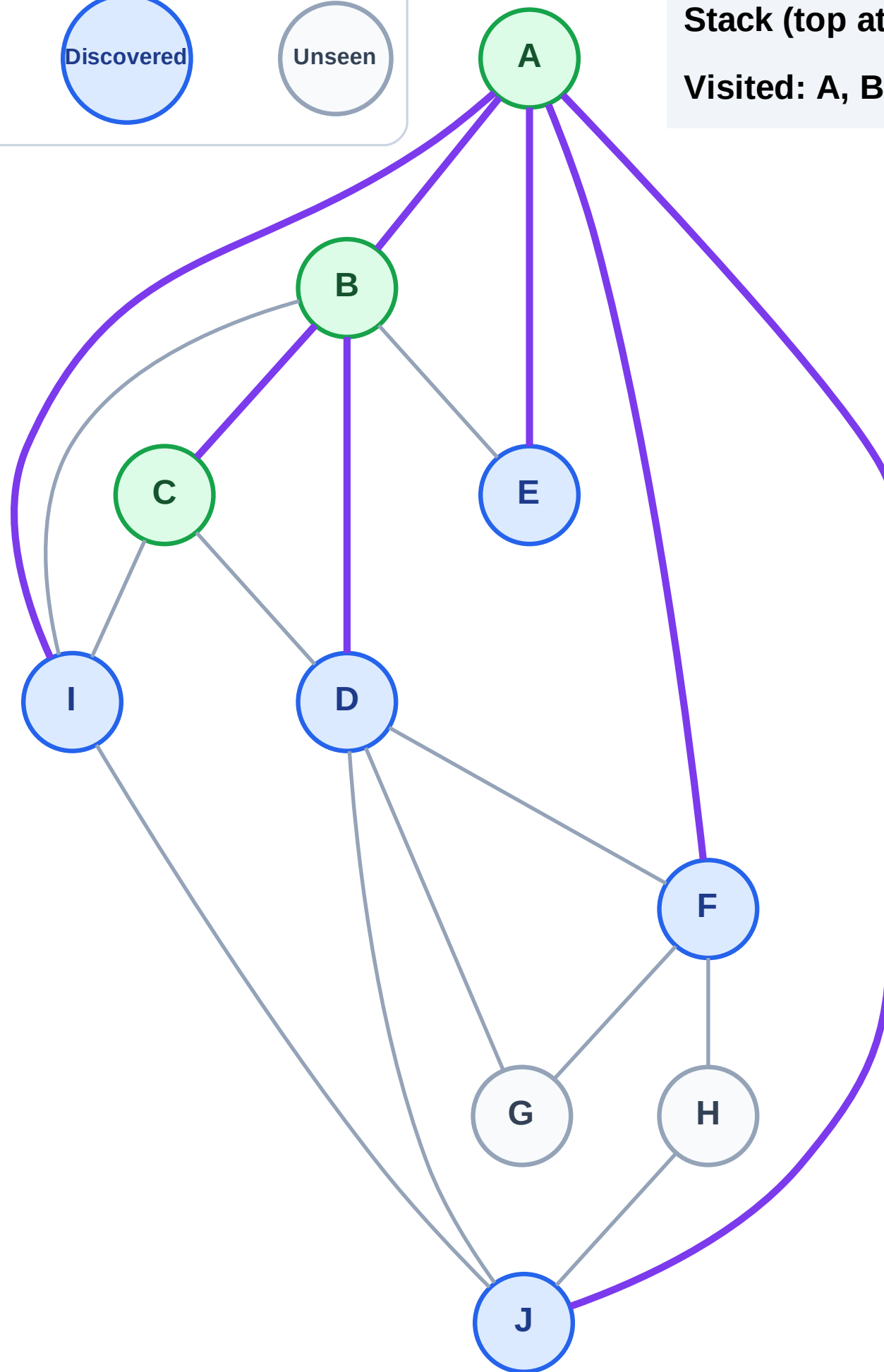
Discovered



Unseen

Stack (top at right): J | I | F | E | D

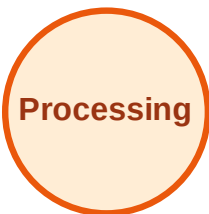
Visited: A, B, C



DFS Traversal

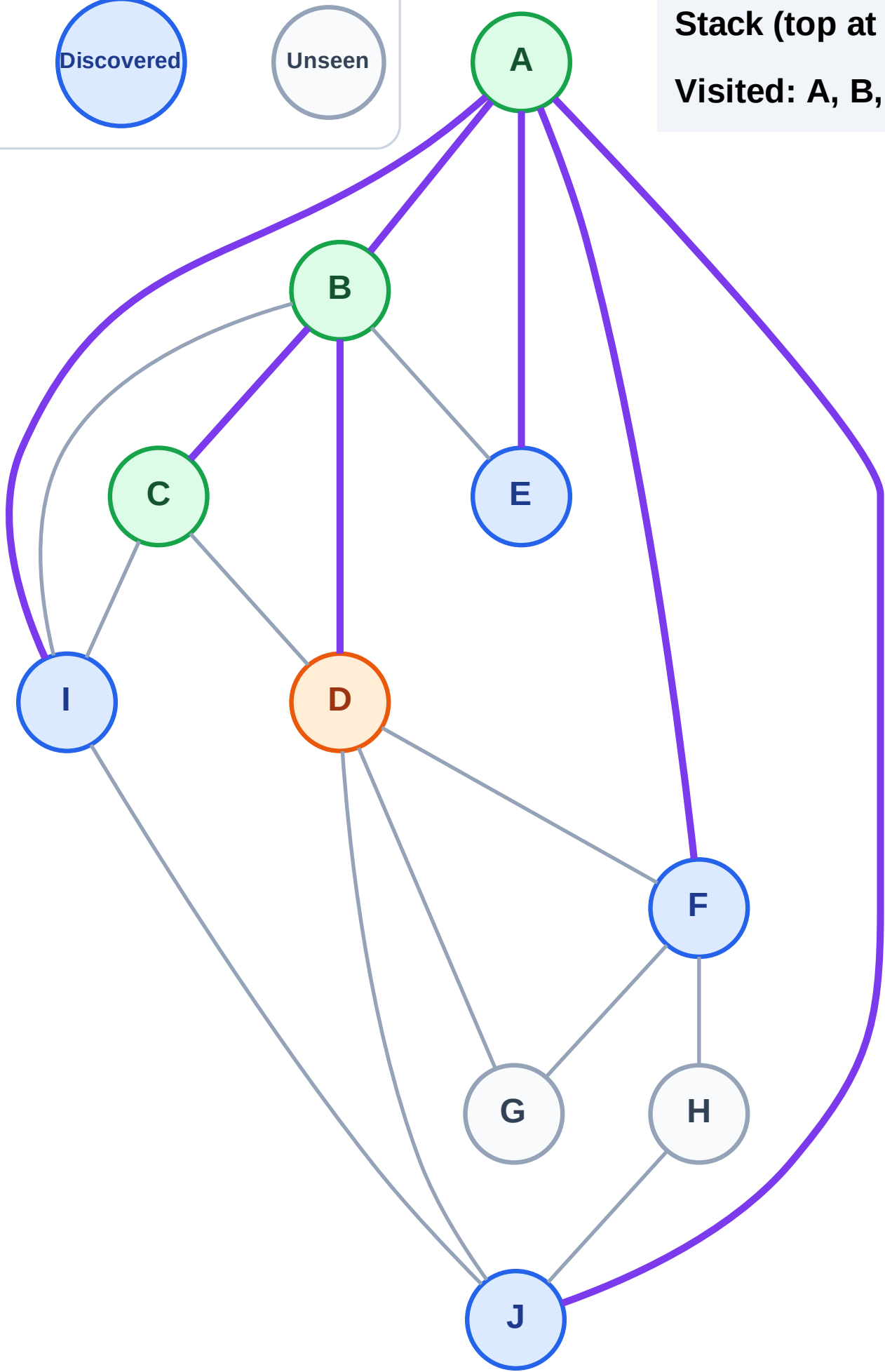
Step 8: Process node D

Legend



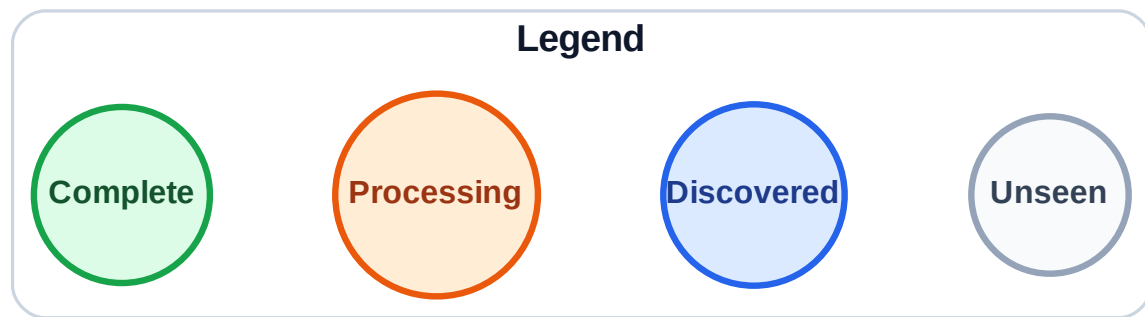
Stack (top at right): J | I | F | E

Visited: A, B, C



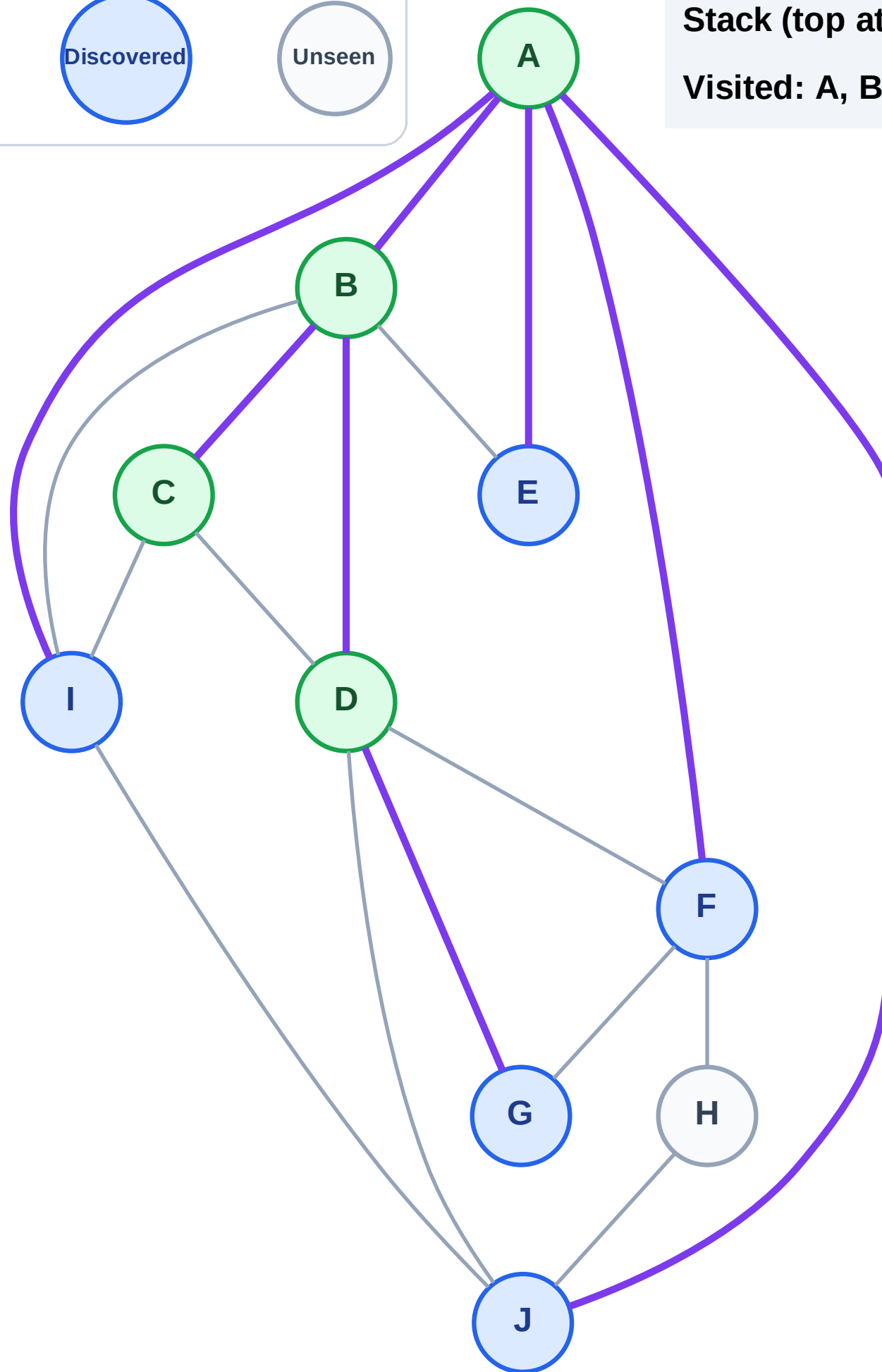
DFS Traversal

Step 9: Discover G from D



Stack (top at right): J | I | F | E | G

Visited: A, B, C, D



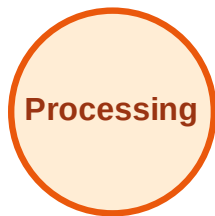
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



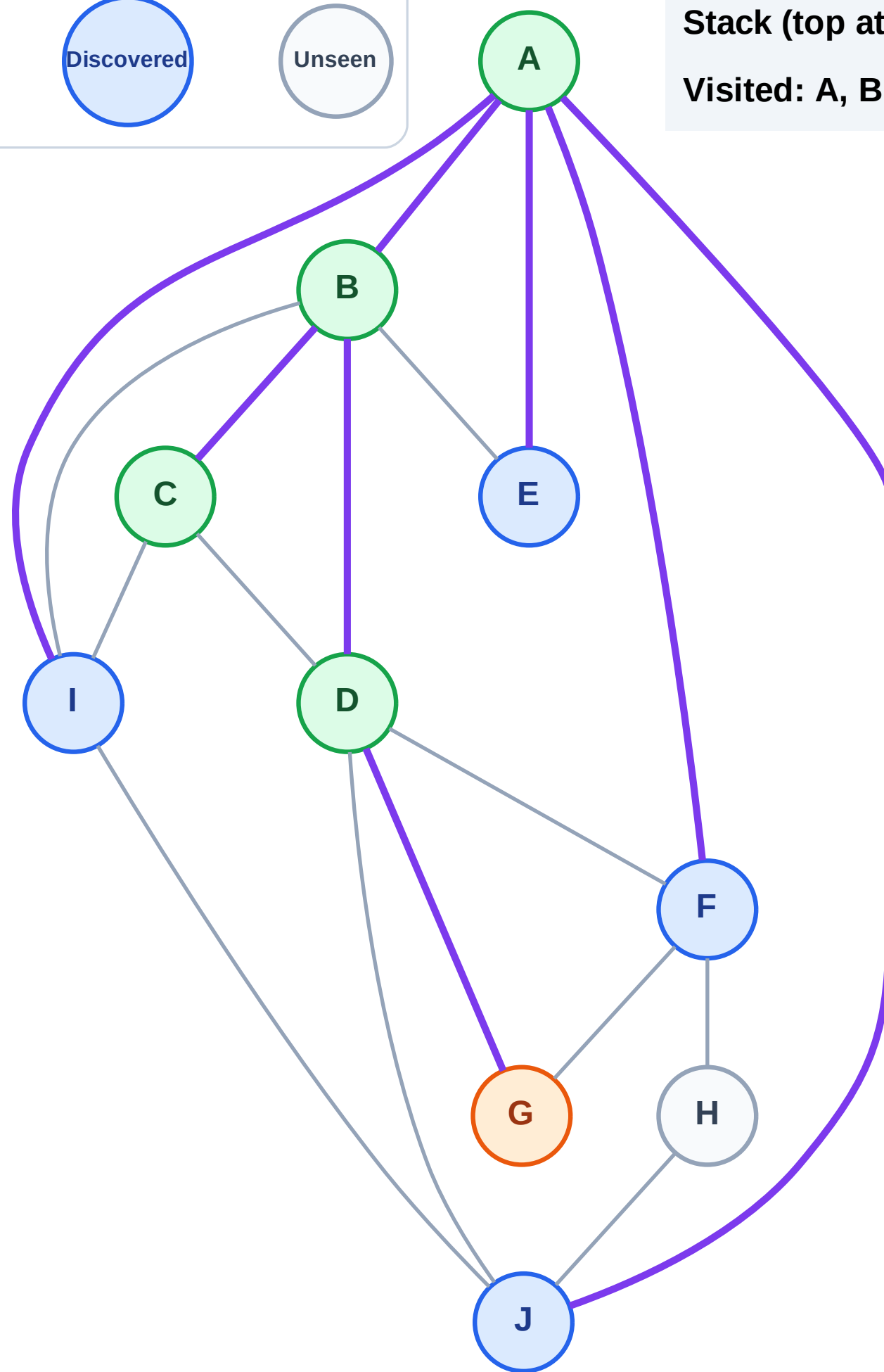
Discovered



Unseen

Stack (top at right): J | I | F | E

Visited: A, B, C, D



DFS Traversal

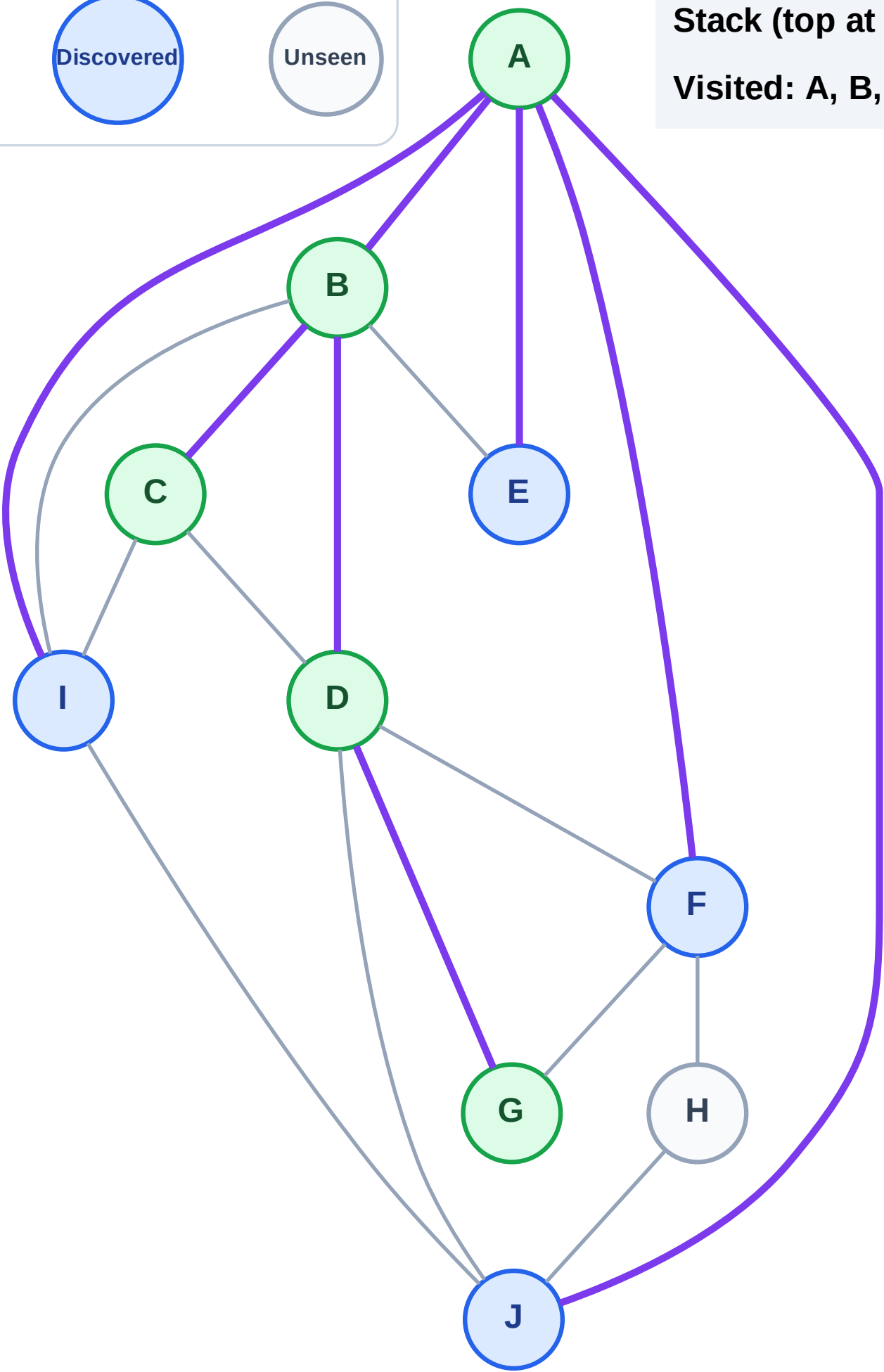
Step 11: No new neighbors from G

Legend



Stack (top at right): J | I | F | E

Visited: A, B, C, D, G



DFS Traversal

Step 12: Process node E

Legend

Complete

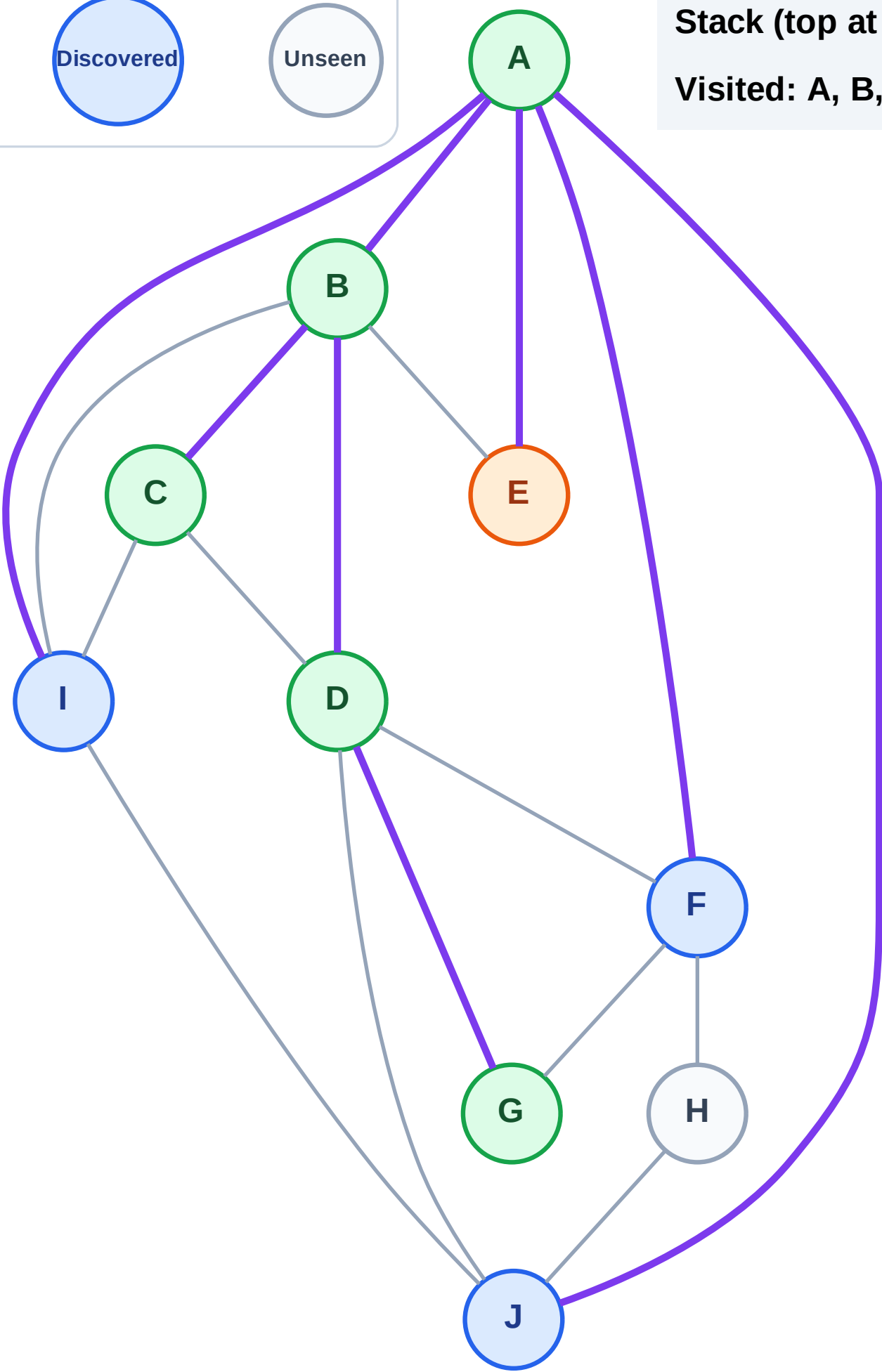
Processing

Discovered

Unseen

Stack (top at right): J | I | F

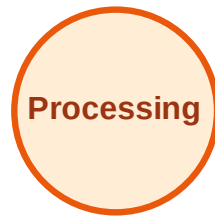
Visited: A, B, C, D, G



DFS Traversal

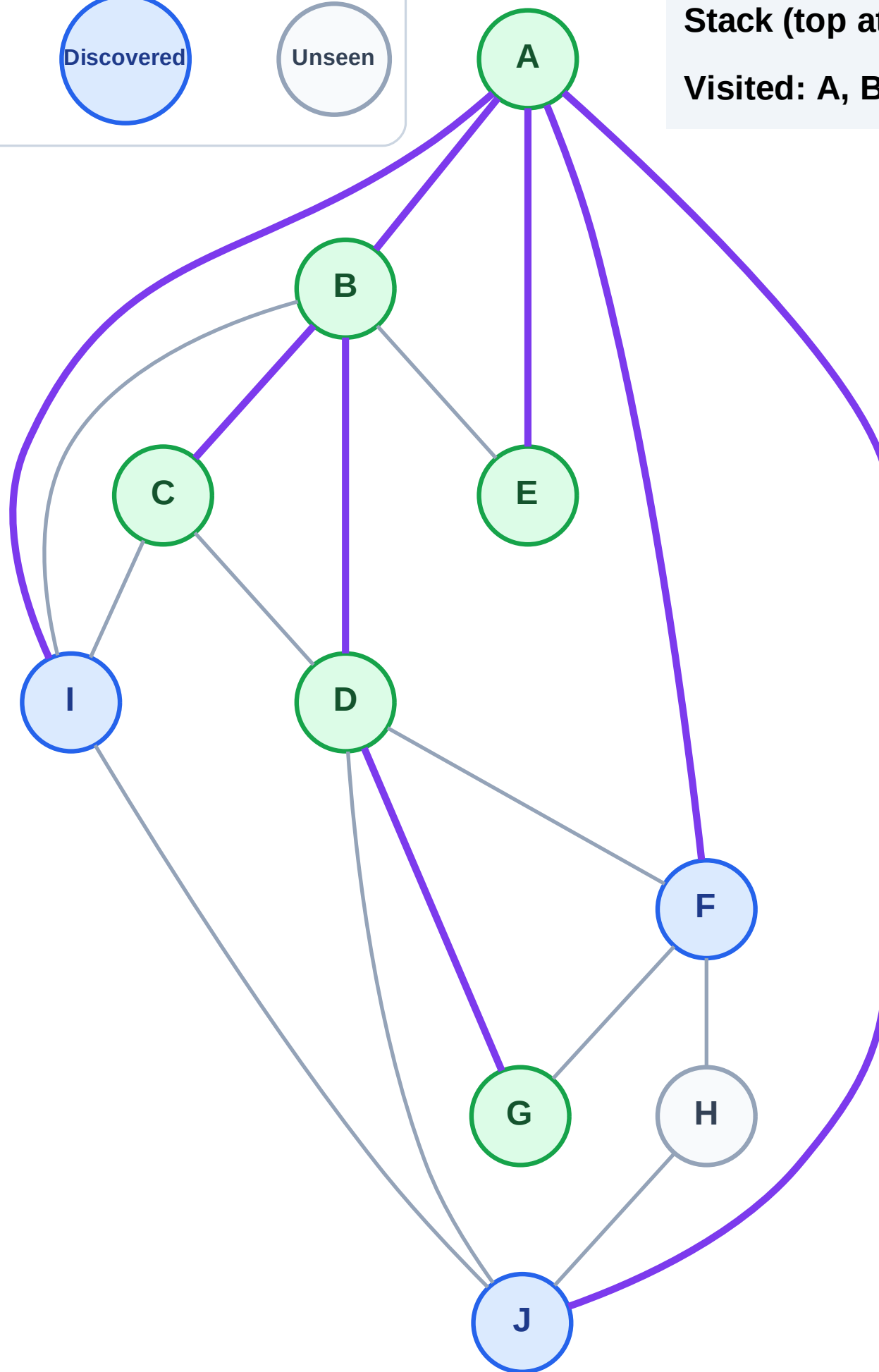
Step 13: No new neighbors from E

Legend



Stack (top at right): J | I | F

Visited: A, B, C, D, E, G



DFS Traversal

Step 14: Process node F

Legend

Complete

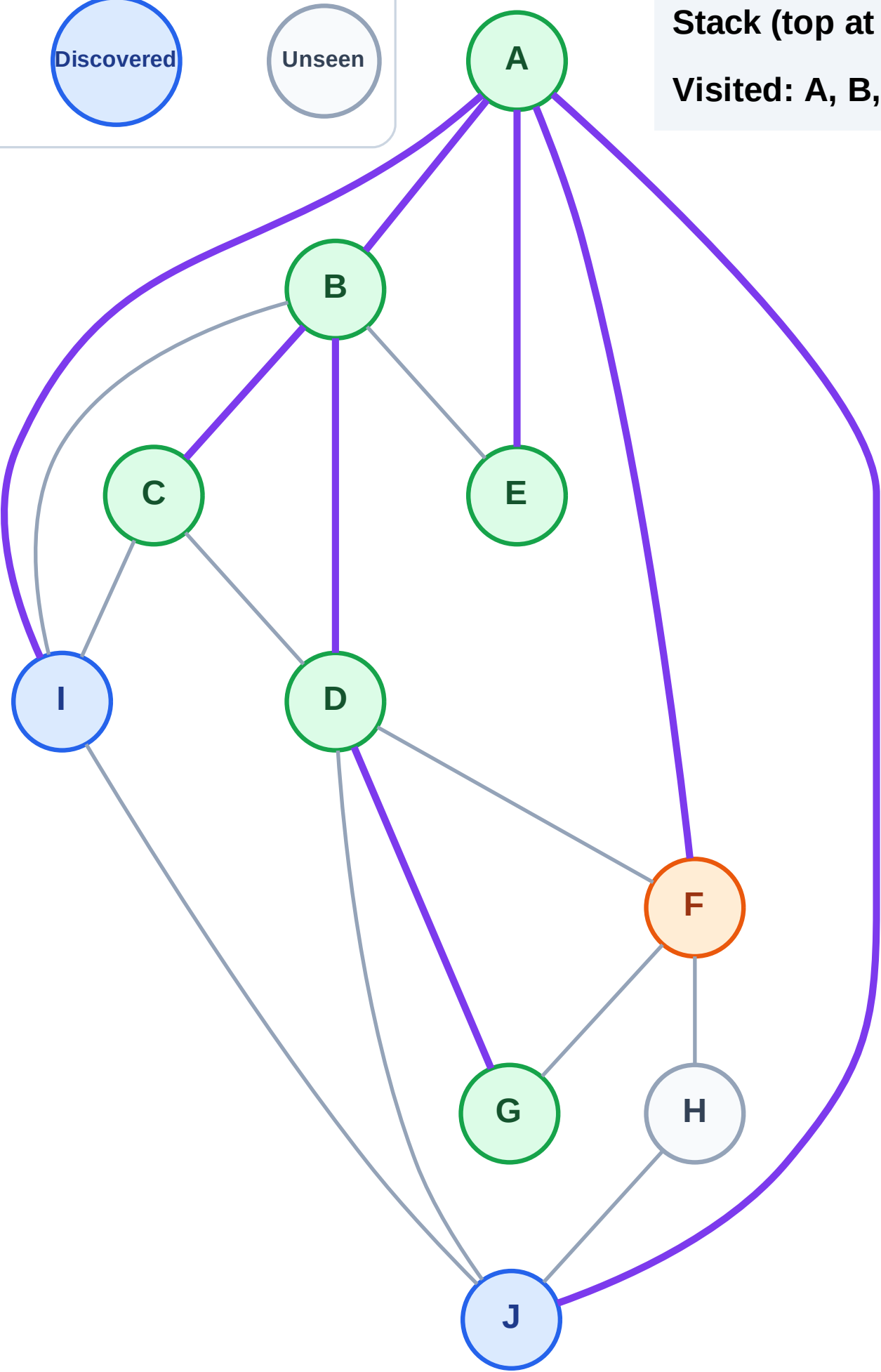
Processing

Discovered

Unseen

Stack (top at right): J | I

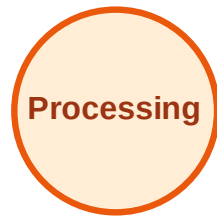
Visited: A, B, C, D, E, G



DFS Traversal

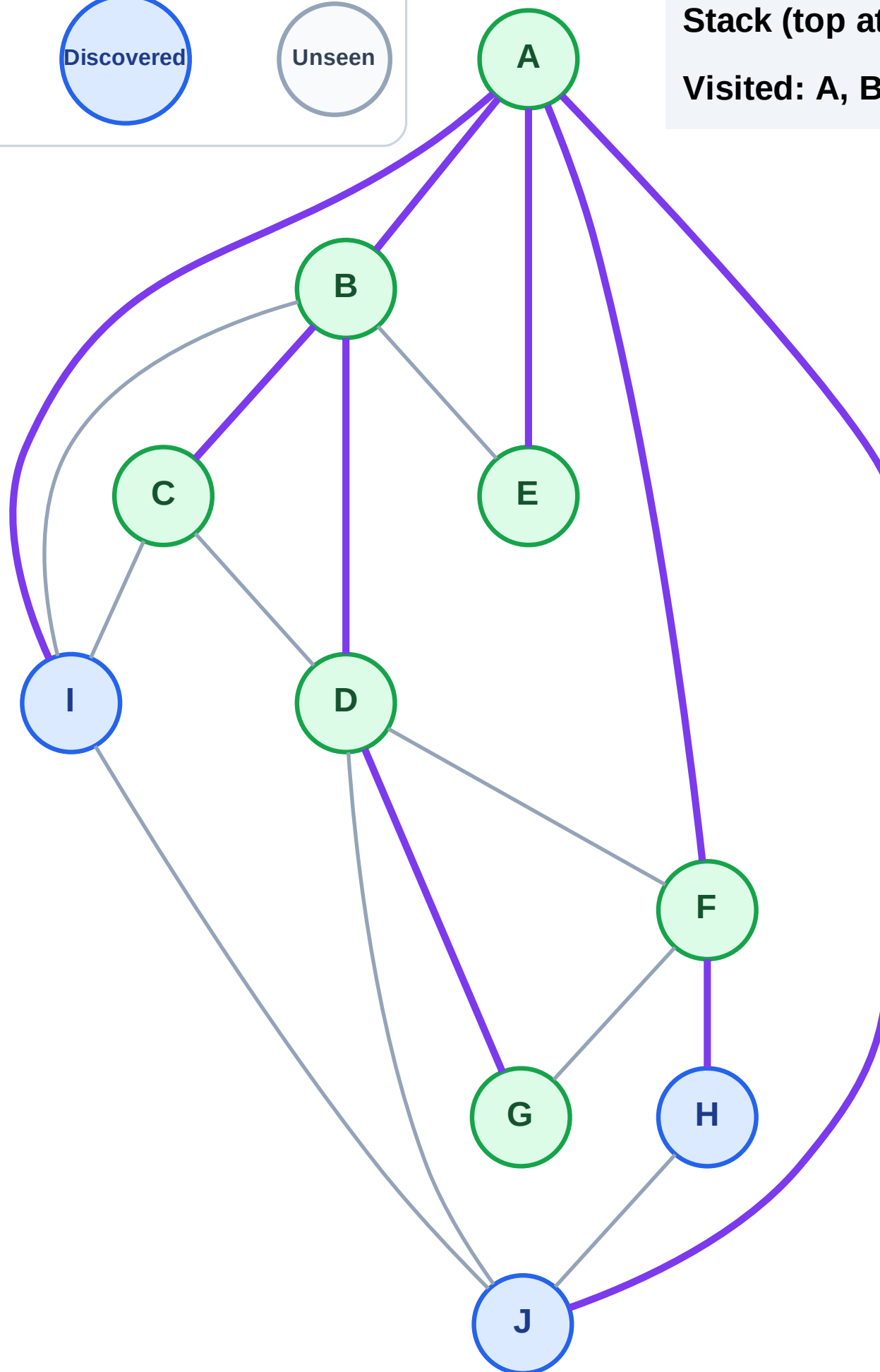
Step 15: Discover H from F

Legend



Stack (top at right): J | I | H

Visited: A, B, C, D, E, F, G



DFS Traversal

Step 16: Process node H

Legend

Complete

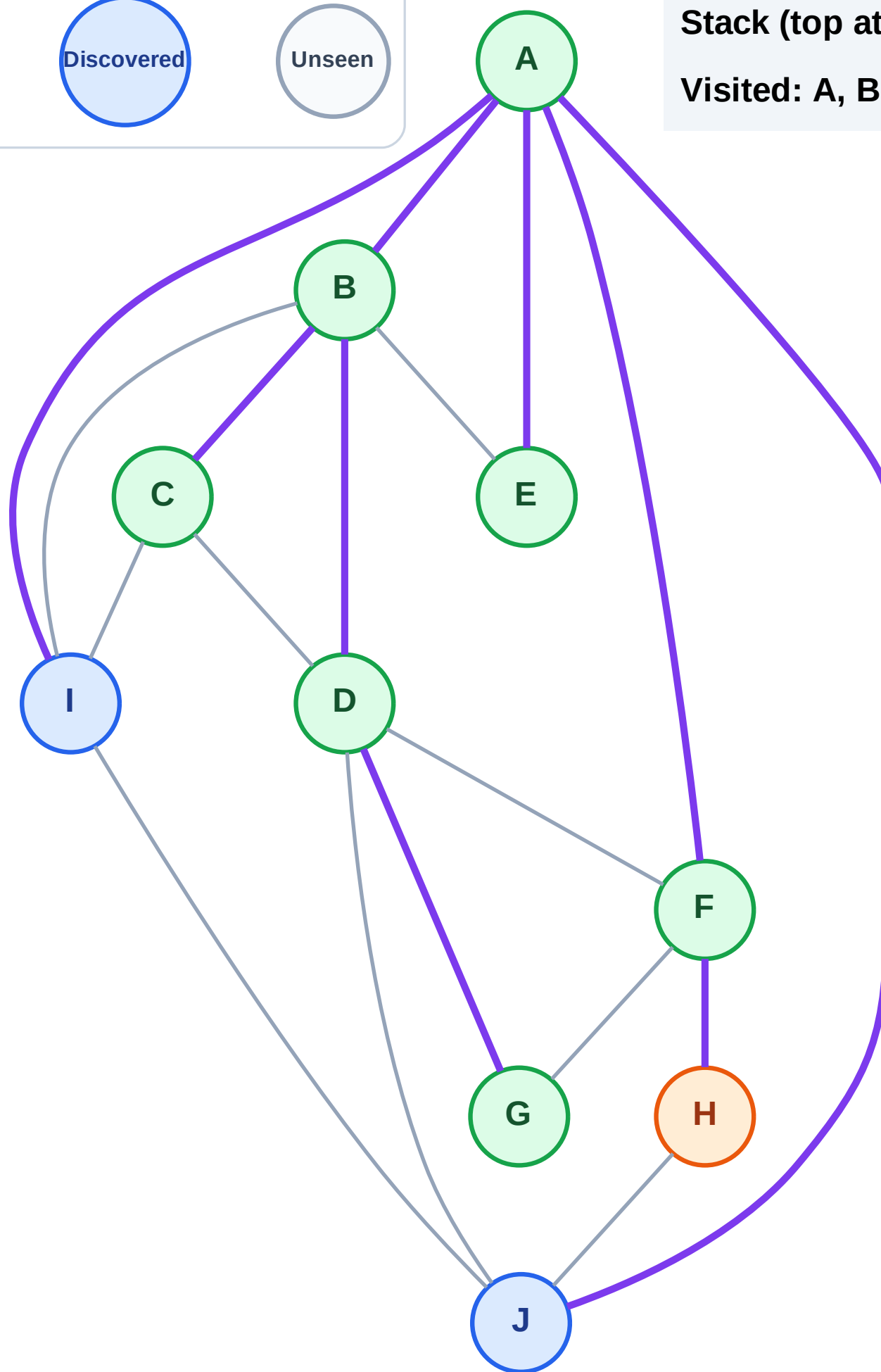
Processing

Discovered

Unseen

Stack (top at right): J | I

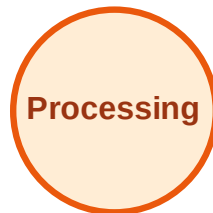
Visited: A, B, C, D, E, F, G



DFS Traversal

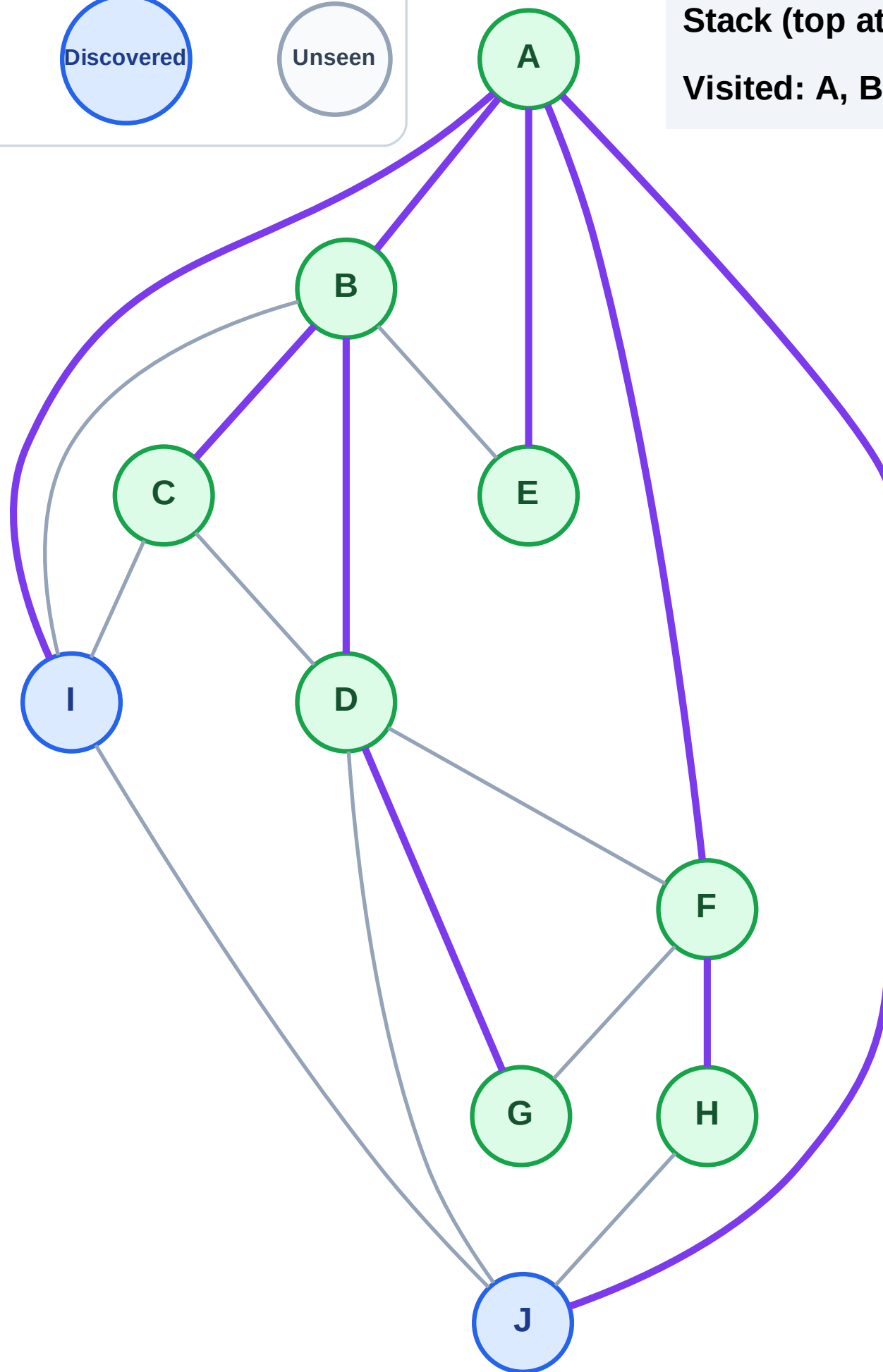
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

Complete

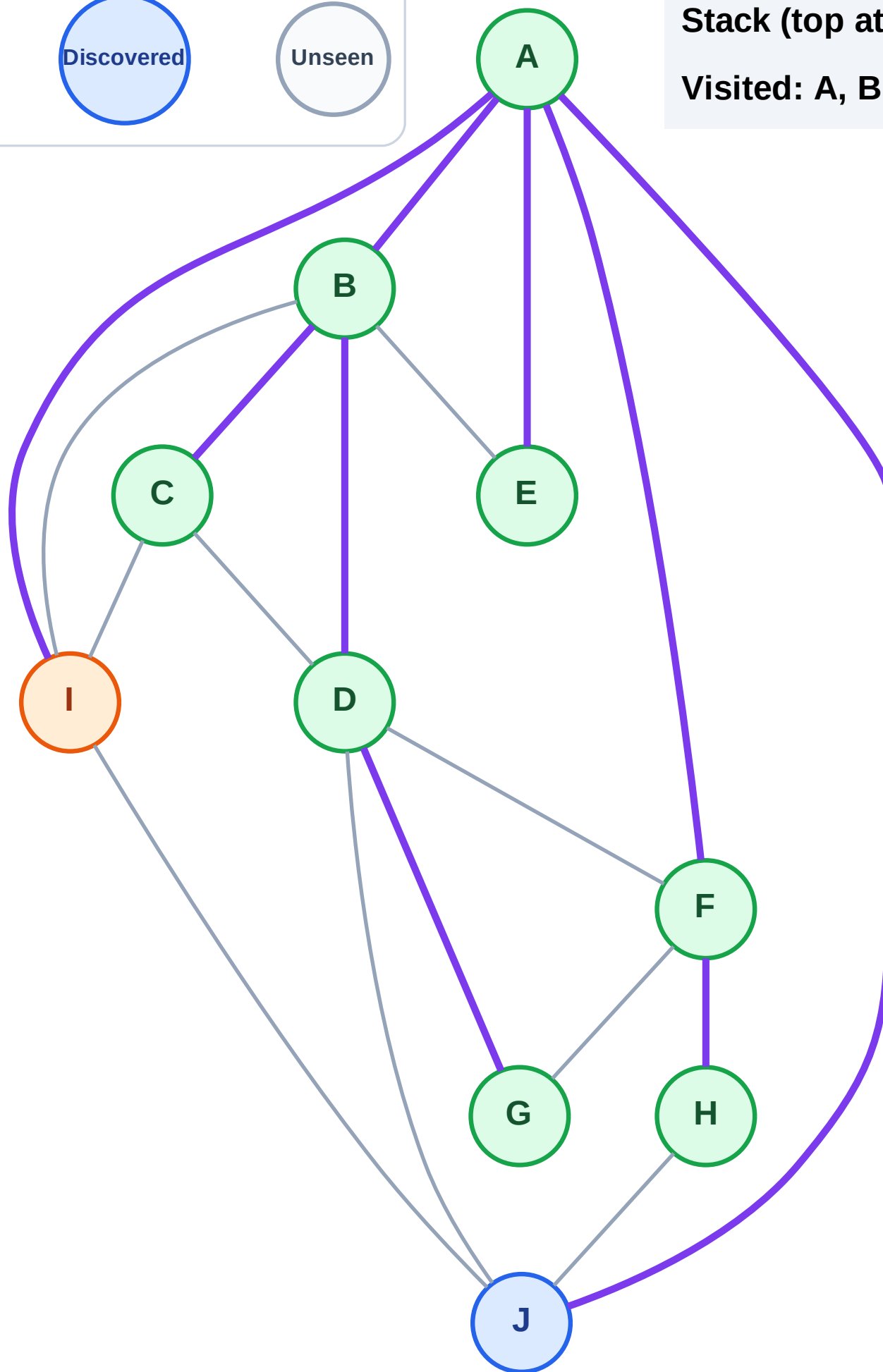
Processing

Discovered

Unseen

Stack (top at right): J

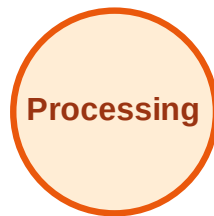
Visited: A, B, C, D, E, F, G, H



DFS Traversal

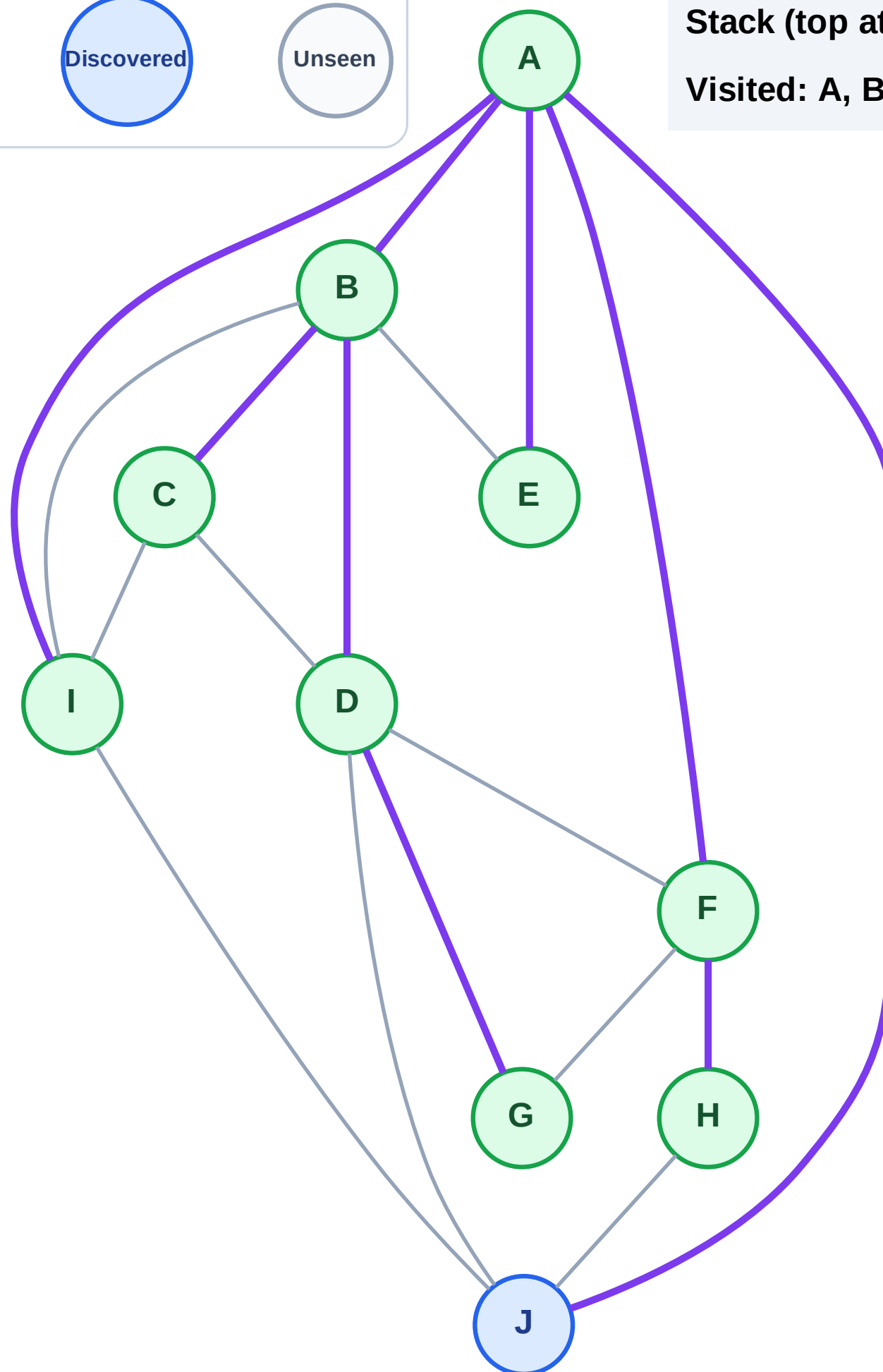
Step 19: No new neighbors from I

Legend



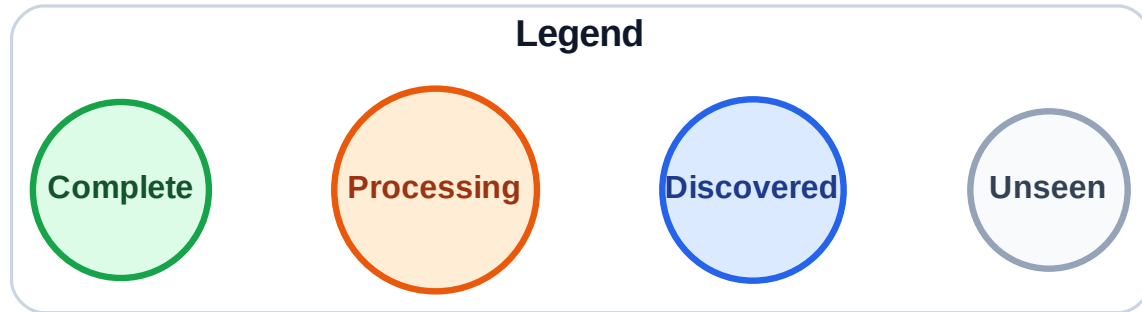
Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



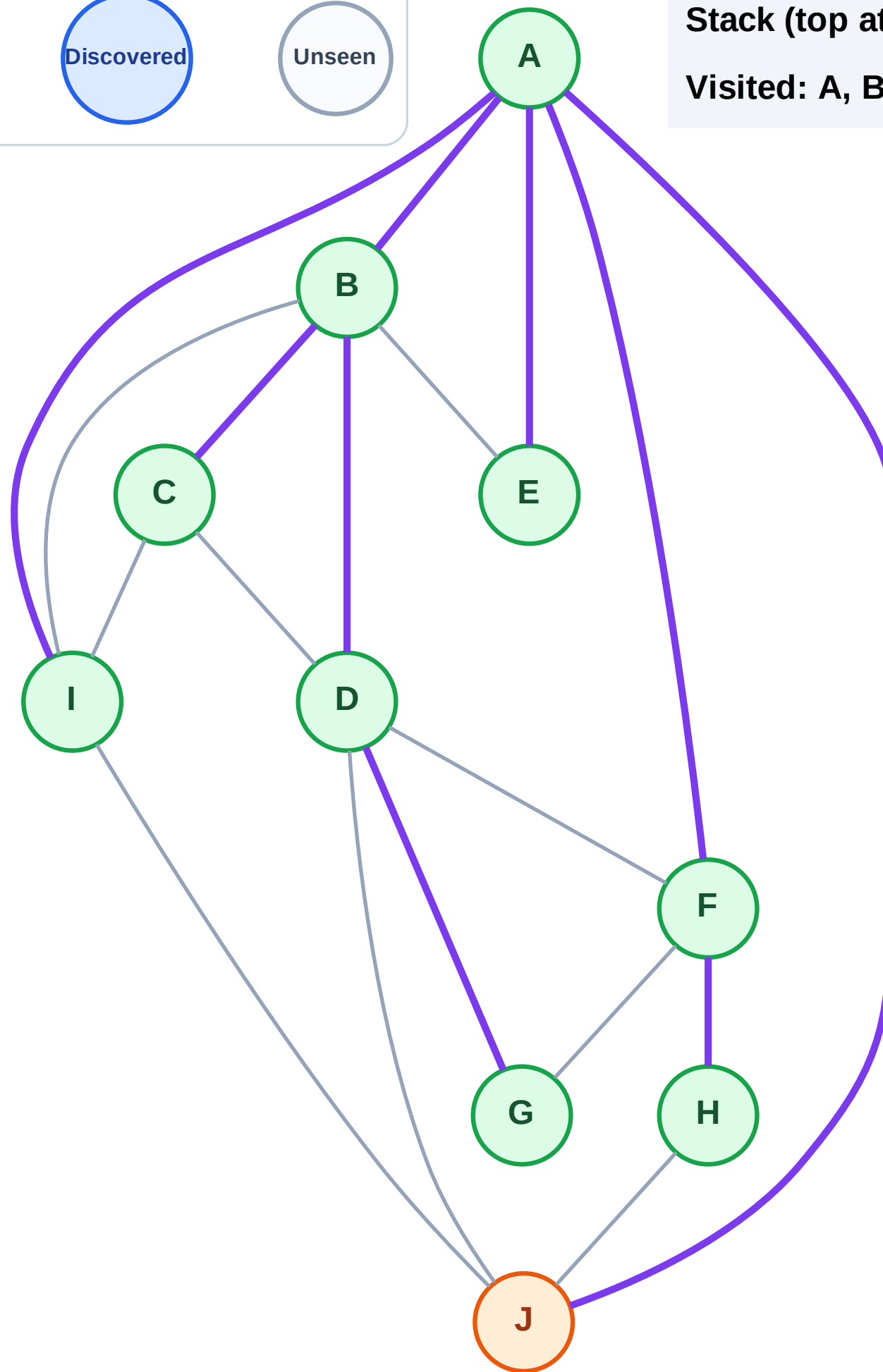
Step 20: Process node J

Step 20: Process node J



Stack (top at right): (empty)

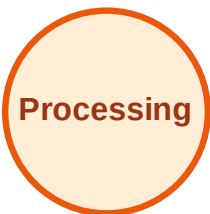
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

